

The Legibility of Income Risk*

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Abstract

Do households perceive the income risk that macroeconomic models assign them? Job stayers in households below \$50k report about 1.8 times the wage uncertainty of those above \$100k, hold these beliefs more persistently, and revise them more strongly following realised wage shocks, while the realised risk estimated from matched same-job spells shows no comparable gradient. Under an estimated learning rule, the bottom group over-perceives its transitory risk, while the middle and top groups under-perceive their permanent risk. I trace both patterns to how legibly pay separates into these two components: the labelled share of transitory income rises from about one-eighth at the bottom to over one-third at the top. In a life-cycle economy, these beliefs reduce liquid wealth by five percent at the bottom and fourteen percent at the top, where households consume over two percentage points more of a windfall within the year. Equating perceived with realised risk therefore overstates precautionary wealth and understates consumption responses, particularly among high-income households.

1 Introduction

How much risk does a worker believe her own wage carries? In the heterogeneous-agent models now standard in macroeconomics, that belief is a primitive of behaviour: an agent who perceives more risk to her future earnings holds a larger precautionary buffer and spends less out of a windfall, whether through prudence or an occasionally binding borrowing constraint. The level and heterogeneity of income risk therefore shape the distribution of wealth, the distribution of marginal propensities to consume, and, through both, the transmission of aggregate shocks and policy. Yet these models rarely measure perceived income risk, and instead simply assign it.

The standard practice is to calibrate income risk indirectly. The econometrician estimates the dispersion of realised earnings in panel data, decomposes it into permanent and transitory components (Meghir and Pistaferri, 2004), and assigns the resulting variances to the agent as the risk she perceives. Perceived risk is thus equated with realised risk: the agent is assumed to know the variances the econometrician recovers and nothing beyond them. The assumption is convenient but strong, and data on subjective expectations now allow it to be tested.

I start by showing that this assumption misses a systematic income gradient in perceived wage risk. The New York Fed's Survey of Consumer Expectations (SCE) elicits from each respondent a density forecast of her own wage growth over the next twelve months, conditional on remaining

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in the same job, with the same hours and employer. Among such workers, those in households earning under \$50k report wage uncertainty about 1.8 times that of workers in households earning above \$100k. They also hold that belief more persistently within person and revise it more strongly following a realised wage shock. None of these gradients appears in the realised wage process: using matched same-job spells in the Survey of Income and Program Participation (SIPP), I find little corresponding income gradient in realised risk, if anything a mild increase with income. The discrepancy is therefore not simply a matter of different groups facing different amounts of wage risk.

The gap becomes clearer when realised risk is decomposed into permanent and transitory components. I find that the bottom group over-perceives its transitory risk, while the middle and top groups under-perceive their permanent risk. Thus, groups lie on opposite sides of their own full-information benchmarks, and total perceived risk crosses the benchmark as well. This two-sided pattern is difficult to reconcile with a single directional source of belief error. I argue instead that it reflects how legibly pay separates into its permanent and transitory components. Higher in the distribution, transitory pay is more often explicitly labelled, for example, as a commission or bonus alongside a regular salary. Workers can therefore net out temporary payments before forming beliefs about persistent earning power, pushing perceived risk downward. At the bottom, pay is more fragmented across variable hours, tips, and irregular supplements. When these components are harder to identify, the resulting measurement noise pushes perceived risk upward.^z

In Section 4, I formalise this idea as a model of belief formation. In the spirit of Pischke (1995), the worker faces a signal-extraction problem: each pay change mixes permanent and transitory shocks, and she must infer the variance of permanent earning power from this noisy signal. Her belief is an exponentially weighted average of past squared surprises, governed by two primitives: the labelled share of pay and the noise with which the remainder is read. She assigns a fixed share of each surprise to the permanent component and does not increase that share when labelling makes the signal cleaner.

The mechanism yields four results. First, greater labelling reduces the weight assigned to each surprise and therefore lowers perceived permanent risk. Second, the perceived transitory prior equals the unlabelled component plus reading noise, so it can lie either above or below its realised counterpart. Third, total perceived risk can consequently lie on either side of realised risk, depending on how legibly pay is delivered. This provides a mechanism for an income gradient in beliefs without an income gradient in the underlying wage process. Fourth, the behavioural consequences depend on which component is misperceived: permanent risk affects precautionary saving, whereas transitory risk has a much weaker saving consequence. The resulting distortion can therefore be largest among high-income workers even though their total perceived risk is the lowest.

In Section 5, I estimate the updating rule within person by income group and find that it organises the three empirical facts. The bottom group updates most slowly, assigning about three times as much of each squared surprise to the permanent component as the top group. Its high level of perceived risk instead comes from a transitory prior almost four times its realised counterpart, while the middle and top groups' transitory priors are at or below their realised levels. I then measure the mechanism's key primitive, the labelled share of within-spell pay movements, using

the same panel that supplies the realised wage process. The labelled share rises from about one-eighth at the bottom to more than one-third at the top. Feeding these measured shares into the model predicts the middle and top groups' prior levels to within a standard error, reproduces the ordering of updating gains, and accounts for roughly half of the estimated income gradient in the response to news. The bottom group's prior remains substantially above the level implied by its labelled share; I interpret the residual as the reading noise generated by fragmented pay. The evidence therefore points to an income gradient in how pay is delivered, rather than in how risky pay is.

Section 6 embeds these measured beliefs in a life-cycle buffer-stock economy in which workers move across income groups as permanent earning power accumulates. Conditional on their income group, workers differ only in the risks they face—the realised wage variances and the job-loss and job-finding rates—while beliefs differ according to the estimates above. Relative to an otherwise identical full-information calibration, the measured beliefs reduce liquid wealth by five percent at the bottom and fourteen percent at the top. The MPC effect is concentrated at the top: these workers hold the most downward-biased beliefs about permanent risk and consume more than two percentage points more of a windfall within the year. The gap in MPCs between the bottom and top therefore narrows by about one-sixth. The bottom group, by contrast, combines a large transitory-risk overestimate with an approximately unbiased permanent-risk belief, leaving its saving behaviour little changed. In one-asset general equilibrium, the belief economy supplies modestly less capital and generates a slightly higher interest rate. Aggregate effects are therefore much smaller than the cross-sectional effects.

The same beliefs also govern the economy's response to changes in risk and pay structure. A reform that raises the labelled share of low-paid work to the level observed at the top reduces precautionary buffers in the newly legible groups by about three percent. Following a persistent rise in permanent risk, the belief economy rebuilds fifteen to nineteen percent less precautionary buffer than the full-information economy over two years because households incorporate only part of the increase into their perceived permanent risk. A full-information calibration therefore overstates the precautionary response to rising permanent risk. By contrast, a recession that changes neither the delivery of pay nor the underlying risk process leaves the two regimes with the same buffer response.

The paper thus makes three contributions. It documents an income gradient in perceived wage risk that is absent from the realised wage process; it shows that this gradient extends to the persistence and updating of beliefs and can place different income groups on opposite sides of their own full-information risk benchmarks; and it proposes and measures a mechanism based on the legibility of pay, with quantitatively important consequences for precautionary saving and the distribution of MPCs.

1.1 Related Literature

The closest paper is Wang (2023), who use the SCE's density forecasts to show that perceived wage risk lies well below the risk conventionally calibrated from panel data, including within demographic groups. They further show that feeding this lower and more heterogeneous risk into an incomplete-market model raises wealth inequality and the share of hand-to-mouth households.

I share his measurement strategy: I use the SCE density forecast to measure perceived risk and compare it with the realised process in the SIPP same-job stayer panel. I also corroborate his central finding: perceived risk lies below the full-information benchmark in every income group, and Section 3 interprets this gap as an upper bound. I depart from his analysis in four respects. First, I organise the gap by income using the matched same-job design and show that, once the realised process is decomposed, the discrepancy is two-sided: the bottom group over-perceives transitory risk, while the middle and top groups under-perceive permanent risk. The income gradient therefore places different groups on opposite sides of their group-specific benchmarks, even though perceived total risk remains below the benchmark for all groups. Second, I document the within-person dynamics of perceived risk, including its persistence and its response to realised shocks, whereas his evidence is cross-sectional. Third, I estimate an explicit updating rule, while he remains agnostic about belief formation by design. Fourth, I propose a microfounded explanation based on how pay is delivered: I measure the labelled share of transitory pay from the itemised pay components in the same records, whereas he attributes the level gap to unobserved heterogeneity or overconfidence. The economic incidence differs accordingly. In his model, belief heterogeneity affects aggregate wealth and its dispersion; here, the wedge is concentrated at the top because the saving-relevant distortion is under-perceived permanent risk, whose gradient is largest at high incomes.

The paper also connects to several strands of the literature on subjective income beliefs. Rozsypal and Schlafmann (2023) document an income-graded bias in expected income levels, whereas I study the second moment. Ben-David et al. (2018) show using the same survey that uncertainty about own-income growth falls with income and education. Koşar and van der Klaauw (2025) characterise the dynamics of the SCE's earnings expectations; I add same-job conditioning, a matched realised process, and an estimated updating rule. Using linked Danish survey and register data, Caplin et al. (2026) likewise find that survey-reported earnings risk is several times below its indirectly estimated counterpart. They attribute most subjective earnings risk to job transitions and find relatively small and symmetric risk conditional on remaining in the same job. Their instrument elicits expectations separately conditional on each possible employment transition, whereas the SCE question studied here conditions on remaining in the same job. Their measure therefore captures the risk associated with employment transitions, while mine isolates the risk that remains within the job. Koşar and Melcangi (2025) estimate how the marginal propensity to consume responds to subjective earnings uncertainty, directly measuring the behavioural margin through which the belief wedge in Section 6 operates. The proposed mechanism also relates to the “insurance versus information” distinction in the partial-insurance literature (Blundell et al., 2008). Pistaferri (2001) and Kaufmann and Pistaferri (2009) use subjective expectations to distinguish anticipated from unanticipated shocks and show that superior information can explain part of the perceived-realised gap. The mechanism here provides a related labelling channel, but one that is observable and varies systematically with income.

The updating rule builds on Pischke (1995), who frames inference about permanent income as a signal-extraction problem. The under-reaction implied by my updating rule is closely related to the information rigidity of Coibion and Gorodnichenko (2015), which Carroll et al. (2020) incorporate into consumption dynamics as sticky expectations. It contrasts with the diagnostic over-reaction

of Bordalo et al. (2018), which Sanchez and Song (2025) embed in an incomplete-markets model. These mechanisms place perceived risk on opposite sides of the full-information benchmark, and the shortfall documented in Section 3 is more consistent with under-reaction. Experience-based learning (Malmendier and Nagel, 2011) provides another source of heterogeneity in beliefs, but its key state variable is age, whereas the gradient documented here rises with income conditional on demographics.

Finally, the proposed mechanism draws on documented differences in how pay is delivered across the income distribution: performance pay is concentrated among high-income workers (Lemieux et al., 2009), high-income earnings shocks are predominantly transitory (Guvenen et al., 2021), while pay is more volatile and fragmented at the bottom (Hardy and Ziliak, 2014; Morduch and Schneider, 2017).

2 Data and Measurement

The paper compares what workers believe about their wage risk with the risk their wages actually carry. I measure perceived risk using the New York Fed’s Survey of Consumer Expectations (SCE) and realised risk using the Survey of Income and Program Participation (SIPP). Both are monthly within-respondent panels, with the SIPP covering reference years 2013–2023 and the SCE covering 2013–2025. I impose the same basic conditioning in both datasets, requiring a worker to remain in the same job with the same employer for at least six consecutive months, and apply survey weights throughout. Respondents are divided into three groups by median annual household income: bottom, below \$50k; middle, \$50k–\$100k; and top, at or above \$50k.¹

The perceived side comes from the SCE density question, which asks each respondent for the distribution of her own earnings growth over the next twelve months “*on the same job, same hours, same employer*”. Responses are reported as a ten-bin histogram, following the density-forecast tradition of Engelberg et al. (2009). This conditioning makes the response useful for isolating wage risk: holding the job, hours, and employer fixed removes predictable changes associated with the worker’s wage path, as well as variation in earnings arising from changes in labour supply. The elicited object is therefore closer to the conditional variance of the stochastic component of wage growth than to total income risk. It also excludes the risk of losing or changing jobs, which constitutes a substantial component of income risk in Caplin et al. (2026); the model in Section 6 reintroduces this margin separately through its employment block. From each reported histogram, I calculate the bin-midpoint variance as the perceived variance $\bar{\sigma}_{\text{perc}}^2$, exclude respondents who place all mass in a single bin, trim at 1%/99%, and drop each respondent’s reports from the first wave in which she records a job change.

The realised side comes from the SIPP, which records earnings monthly and provides a unique employer identifier, allowing me to impose the same-job and same-employer conditions at the frequency targeted by the SCE question. Monthly earnings comprise regular wage and salary income together with bonuses, commissions, tips, and overtime, recorded as the amounts actually

¹The cut-offs are nominal and fixed across the sample. The SCE skews towards higher incomes, so group sizes are uneven (estimation sample: 1,531 / 7,501 / 11,331 observations). The \$50k cut-off is chosen because finer income brackets become too thin to support the estimation (Appendix E.1), although a finer partition could locate more precisely where the gradient emerges.

paid; I exclude the self-employed. Unlike the SCE measure, these earnings also vary with hours actually worked. In Appendix C.7, I isolate the contribution of this difference in conditioning and find that accounting for it strengthens rather than overturns the main findings. I drop January observations, where reported cross-wave changes bunch spuriously, and trim squared changes at 1%/99%.

I residualise both objects on standard demographic controls before computing any moments, following Wang (2023): age and its square, sex, education, state, and year-month fixed effects, with the latter absorbing aggregate shocks common across groups. On the perceived side, this yields a residualised measure of perceived variance, $\bar{\sigma}_{\text{perc}}^2$; on the realised side, it yields a residualised log wage. The residualisation removes systematic variation predictable from demographics, location, and aggregate time effects, leaving the stochastic component on which the comparison is based. Finally, the SCE specification includes a survey-tenure term to absorb the panel conditioning associated with repeated subjective reports (Kim and Binder, 2023); no corresponding term is included on the realised-pay side.

3 Perceived and Realised Risk

This section sets out the facts the rest of the paper must explain. Perceived risk is steeply graded by income, and so is its persistence within person. Neither gradient has a counterpart in the realised wage process, which I estimate on matched same-job spells in the second half of the section. Realised wage shocks do, however, move beliefs over time.

3.1 Levels

Perceived variance is steeply graded by income, with the bottom of the distribution far above a middle and top that sit together. As Figure 1 shows, the weighted group means are 20.6 at the bottom, 11.3 in the middle, and 11.4 at the top. A worker at the bottom thus perceives about 1.8 times the wage uncertainty of a worker at the top. Ben-David et al. (2018) document in the same survey that uncertainty about own income growth falls with income and education. Here, the matched-stayer conditioning and the realised counterpart below sharpen that pattern into a statement about wage risk.

The gradient takes the form of a step: the middle and top groups are statistically indistinguishable under the midpoint measure and track each other closely, while the gap between the bottom and the other two groups survives changes in the income cut-off. The gradient also survives conditioning on numeracy and inflation uncertainty, both of which are themselves steeply graded by income (Appendix C.4). The bottom series drifts upward over the sample, which may reflect the fixed nominal cut-offs or the greater cyclical exposure of low-income pay.

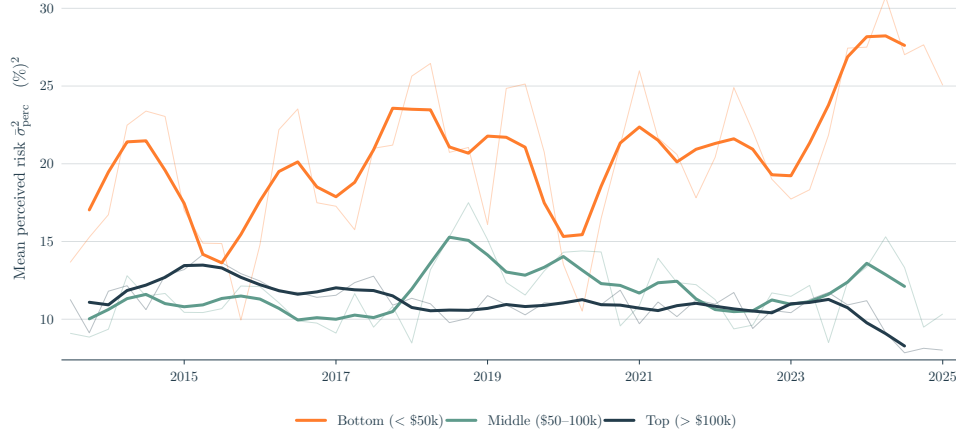


Figure 1: Group-mean perceived variance $\bar{\sigma}_{\text{perc}}^2$ by income group: quarterly means (thin) and their 4-quarter centred moving average (thick). Matched-stayer, trimmed and residualised SCE panel as in Section 2.

3.2 Persistence

Perceived risk is also more persistent within person at the bottom than at the top. I estimate a within-respondent AR(1) in perceived variance $\bar{\sigma}_{\text{perc}}^2$, with persistence coefficient $\hat{\phi}_g$, using Blundell–Bond system GMM (Blundell and Bond, 1998). The short, wide SCE panel makes the within estimator severely Nickell-biased, so I instrument the lagged dependent variable with deeper lags to address this bias, following Ma et al. (2020), who uses the same estimator for a short-panel setting.

Group	Monthly		Quarterly	
	$\hat{\phi}_g$	SE	$\hat{\phi}_g$	SE
Bot	+0.343***	0.046	+0.632***	0.062
Mid	+0.142***	0.026	+0.594***	0.088
Top	+0.144***	0.025	+0.377***	0.076
Wald ($B = T$)	$p = 0.0001$		$p = 0.009$	

Table 1: Within-respondent AR(1) persistence $\hat{\phi}_g$ of perceived variance, Blundell-Bond system GMM. Instruments: $\bar{\sigma}_{\text{perc}}^2$ at lags 3–4 in differences plus the BB level moment, collapsed. Two-way clustered SEs (respondent, quarter). The same instrument set is used at both cadences.

The bottom group is the most persistent of the three, and equality with the top group is rejected at both monthly and quarterly cadences. This gradient does not appear to be inherited from the realised process: in Appendix C.5, I show that realised volatility instead becomes more persistent with income. Nor is the gradient an artefact of the reporting cadence. Classical measurement noise in the monthly reports biases monthly persistence estimates downward and is partly averaged out at the quarterly frequency, pushing the quarterly estimates upward. Noise alone therefore cannot account for the gap, since doing so would require an implausibly persistent underlying process. At either cadence, the bottom group holds its beliefs for longer than the top group.

Taken together with the level differences, these persistence estimates imply that calibrating per-

ceived income risk to realised wage data requires reproducing not only the income gradient in the level of perceived risk, but also the gradient in how persistently workers hold those beliefs.

3.3 The Realised Wage Process

To compare realised and perceived risk at the survey's horizon, I estimate the standard permanent-transitory decomposition of Meghir and Pistaferri (2004) on matched same-job spells:

$$w_{i,t} = z_{i,t} + e_{i,t}, \quad e_{i,t} = p_{i,t} + \theta_{i,t}, \quad p_{i,t} = p_{i,t-1} + \psi_{i,t}. \quad (1)$$

Here $w_{i,t}$ is the residualised log wage, $z_{i,t}$ the predictable Mincerian component, and $e_{i,t}$ the stochastic component that is the focus of the rest of the paper. The permanent shock $\psi_{i,t}$ is iid with variance $\sigma_{\psi,g}^2$, where g indexes the worker's income group, and accumulates through the random walk $p_{i,t}$. Economically, it captures shocks such as merit raises or cost-of-living adjustments. The transitory shock $\theta_{i,t}$ is iid with variance $\sigma_{\theta,g}^2$ and captures temporary fluctuations such as discretionary bonuses or unpredictable working hours.

The SCE question targets this stochastic component. Because it conditions on *the same job, hours, and employer*, the predictable component is held fixed, so the elicited variance corresponds to

$$\bar{\sigma}_{\text{perc},i,t}^2 \equiv \text{Var}_{i,t}(\Delta w_{i,t+1} \mid \Delta z = 0) = \text{Var}_{i,t}(\Delta e_{i,t+1}). \quad (2)$$

On the SIPP side, I residualise wages on demographics to remove z and then difference the residual to recover Δe directly. The recorded wage is monthly earnings per average weekly hour reported for the month, so it includes tips, overtime, and premium pay. Both measures therefore target the stochastic component of wage changes within the same employer, with the qualification that the SCE also conditions on hours.

For the baseline realised moments, I retain all months, whether or not reported hours change. A persistent change in scheduled hours is itself a source of income risk borne by the worker, even though the SCE question conditions it away. I examine the alternative conditioning on unchanged hours separately in Appendix C.7.

Two moments per income group identify the two wage primitives. The one-period change is

$$\Delta e_t = \psi_t + \theta_t - \theta_{t-1}, \quad (3)$$

and let $\bar{S}_{m,g} \equiv \mathbb{E}[(\Delta e)^2]$ denote its variance. Because ψ and θ are independent and iid,

$$\bar{S}_{m,g} = \sigma_{\psi,g}^2 + 2\sigma_{\theta,g}^2, \quad \text{Cov}(\Delta e_t, \Delta e_{t-1}) = -\sigma_{\theta,g}^2. \quad (4)$$

The first-order autocovariance therefore identifies the transitory variance, while the remaining variance identifies the permanent variance, separately for each income group. Autocovariances beyond the first lag are close to the zero restrictions implied by the model, and conditioning on stayers does not materially change the estimates (Appendix C.6).

Group	$\sigma_{\theta,g}^2$ (transitory)	$\sigma_{\psi,g}^2$ (permanent)	$\bar{S}_{m,g}$
Bot	2.52	11.89	16.92
Mid	4.42	11.60	20.44
Top	5.83	11.22	22.88

Table 2: SIPP permanent-transitory identification by income group, monthly cadence, in (%)². Moments are computed from unrounded inputs and rounded for display.

The realised gradient runs, if anything, in the opposite direction to the perceived gradient (Table 2). Permanent variance is approximately flat across income groups, while transitory variance rises with income, plausibly reflecting the greater importance of bonus pay among higher-income workers. Conditioning on unchanged reported hours sharpens rather than overturns this pattern: both components then rise mildly with income (Appendix C.7). The contrast with the perceived gradient therefore does not depend on the treatment of hours.

3.4 The FIRE Benchmark

The perceived and realised objects meet at the twelve-month horizon targeted by the survey. Rolling the monthly process forward twelve months under iid shocks (derivation in Appendix A) gives

$$\text{Var}(e_{t+12} - e_t) = 12\sigma_{\psi,g}^2 + 2\sigma_{\theta,g}^2, \quad (5)$$

which is the twelve-month risk a full-information rational-expectations (FIRE) agent would report. This benchmark is around 148 (%)² for every group, and falls to 79–98 under the same-hours conditioning in Appendix C.7. Perceived variance is substantially below either benchmark, at 20.6, 11.3, and 11.4, corresponding to between one-quarter and one-thirteenth of the benchmark (Table 4, Appendix A). The gap between perceived and realised income risk is similar in magnitude to that documented by Wang (2023).

How much of this gap reflects genuine miscalibration is difficult to say, because the benchmark itself is fragile. Equation (5) multiplies the permanent variance by twelve, while SIPP earnings are self-reported: the classical component of reporting error is treated as transitory, but any persistent component is indistinguishable from true permanent shocks and therefore enters the benchmark twelvefold (Bound and Krueger, 1991). Administrative payroll data, which are free from reporting error, place job-stayers' base-wage dispersion well below the benchmark (Grigsby et al., 2021). Reporting error is plausibly common across groups, however, so its effect should largely cancel in the cross-group comparison. I therefore interpret the level gap only as an upper bound and base the argument primarily on the contrast across groups.

3.5 Belief Dynamics and Realised Shocks

The cross-section rules out the realised process as an explanation for the income gradient, but realised shocks can still shape belief dynamics. Figure 2 shows that pooled perceived risk co-moves with the absolute deviation of aggregate wage growth from its sample mean. The pattern

is consistent with wage-growth surprises raising perceived risk over time.²



Figure 2: SCE perceived risk and the absolute deviation of SIPP aggregate wage growth from its sample mean, both standardised and pooled across income groups, 2015–2022.

Together, these facts impose two requirements on any belief rule. Perceived variance must be allowed to move over time, since the dynamics rule out constant beliefs, and realised shocks must be able to feed into those movements. Whether the response to a largely common signal differs systematically across income groups is an empirical question, which I take up using within-person variation in Section 5.

4 Belief Formation under Illegible Pay

This section develops a model of how workers infer income risk from the pay they observe, and what this implies for saving. The worker sees one pay change each period, which combines permanent and transitory shocks, and must infer how much of the variation reflects a lasting change in earning power. Some transitory pay arrives labelled, for example as a bonus or commission on a separate payslip line, and the worker can *net* it out before forming her belief. The remainder mixes permanent and transitory variation, and she cannot decompose it any further.

The mechanism generates four results. First, labelling makes the signal about permanent earning power cleaner, but the worker does not fully adjust her attribution rule to account for this improvement. She therefore assigns too little of each observed surprise to the permanent component, and her perceived permanent risk falls below the true variance. Second, the same netting reduces the transitory variation left in the signal, pushing perceived transitory risk downward. Third, noise in reading the unlabelled remainder is itself indistinguishable from a transitory shock and therefore raises perceived transitory risk. Total perceived risk can consequently lie either above or below realised risk: the permanent component is always under-perceived, while the transitory component can be either over- or under-perceived. Fourth, the two errors have different behavioural consequences. Permanent-risk misperception affects precautionary saving, whereas transitory-

²I use absolute deviations because a realised shock of either sign should increase perceived uncertainty; squared deviations produce a very similar pattern. The comovement motivates the shock regressor used in Section 5, where the within-person estimates provide the evidence on whether realised shocks move beliefs.

risk misperception is largely buffered within the period. The saving distortion therefore need not be largest for the worker with the largest total belief error.

Two primitives govern the mechanism. The first is pay legibility, ρ_g , the share of transitory pay that arrives sufficiently identified for the worker to net it out before updating. The second is reading noise, ω_g^2 , the variance of the error with which she interprets the remaining pay. I treat both as primitives of the belief process and measure them in Section 5. The labelled share is constructed from the itemised components of the same pay records used to estimate the realised wage process, while reading noise is identified from the component of the transitory prior left unexplained by the measured share.

4.1 Environment

A worker's wage follows the permanent-transitory process of Section 3, $\Delta e_t = \psi_t + \theta_t - \theta_{t-1}$, where ψ_t is an iid permanent shock with variance $\sigma_{\psi,g}^2$ that accumulates in a random walk, and θ_t is an iid transitory shock with variance $\sigma_{\theta,g}^2$ that disappears within the month. The worker knows the form of this process but must learn the variance of permanent earning power from the pay she observes.

The worker observes one pay change per period. Since permanent and transitory shocks enter the same pay change, the raw squared change contains both components: $\mathbb{E}[(\Delta e)^2] = \sigma_{\psi,g}^2 + 2\sigma_{\theta,g}^2 \equiv \bar{S}_{m,g}$. Her problem is therefore a signal-extraction problem in the spirit of Pischke (1995). She is not trying to infer the level of permanent income from noisy observations, but rather the variance of the permanent component from observations that also contain transitory variation. A household may therefore track the level of its persistent income well, as Druedahl and Jørgensen (2020) infer from consumption responses, while still misperceiving the variance of its income shocks.

Pay is not necessarily observed as one undifferentiated number. Let $\rho_g \in [0, 1]$ denote the share of transitory pay that arrives labelled as transitory before the worker updates. A labelled payment might appear as a separate line on a payslip, be announced in advance with its amount, or recur at a predictable time of year. The worker can therefore net the labelled component out of the pay change and update only on what remains. Labelling is plausibly more common higher in the income distribution, where performance pay is often explicitly identified (Lemieux et al., 2009). It can also arise through other institutional arrangements. The measure used in Section 5 captures only labelling that is visible in the itemised pay records, so it should be interpreted as a lower bound on the information available to workers.

For intuition, consider a salaried worker whose pay is a few hundred euros higher in one month because of a commission. If the commission appears as a separate line on the payslip, she can identify the source of the increase and net it out. The remaining salary change is close to zero, so the observation contains little news about her permanent earning power. By contrast, when the same increase arrives through a combination of variable hours, tips, and irregular supplements, the worker may not be able to identify which part of the change is transitory.

Write the transitory shock as $\theta_t = \theta_t^l + \theta_t^u$, where the labelled and unlabelled components have variances $\text{Var}(\theta_t^l) = \rho_g \sigma_{\theta,g}^2$ and $\text{Var}(\theta_t^u) = (1 - \rho_g) \sigma_{\theta,g}^2$. After netting out the labelled component,

the residual signal has variance

$$V_{u,g} = \sigma_{\psi,g}^2 + 2(1 - \rho_g)\sigma_{\theta,g}^2 \leq \bar{S}_{m,g}, \quad (6)$$

with equality when no transitory pay is labelled. Greater legibility therefore reduces the variance of the signal the worker updates on while leaving its permanent component unchanged. The residual is therefore a cleaner signal of permanent earning power.

The second primitive is the worker's ability to read the unlabelled remainder. Consider a worker whose pay varies with shift counts, cash tips, and irregular supplements, potentially alongside changes in hours (Hardy and Ziliak, 2014; Morduch and Schneider, 2017). She may observe that monthly pay has increased without being able to determine precisely how much of the increase reflects her wage rather than other components of earnings. In Appendix D.5 I show that within-person hours volatility indeed falls monotonically with income.

I capture this uncertainty as reading noise:

$$\tilde{y}_t = p_t + \theta_t^u + v_t, \quad v_t \stackrel{\text{iid}}{\sim} (0, \omega_g^2), \quad (7)$$

where p_t is the permanent component and v_t is the worker's reading error. The error enters the observed change as Δv_t . It therefore has variance $\text{Var}(\Delta v_t) = 2\omega_g^2$, and first-order autocovariance $\text{Cov}(\Delta v_t, \Delta v_{t-1}) = -\omega_g^2$, with zero autocovariance at higher lags. Its statistical signature is therefore identical to that of a transitory shock. The worker cannot distinguish the two from her own observations, so reading noise is recorded as transitory risk in her beliefs. I am not taking a stand on specifying the source of this error since misreading pay components, uncertainty about hours, and differences in reporting or interpretation all generate the same object in the worker's information set.

4.2 Modelling Choices

The survey provides one perceived-risk series per worker, while the model contains two latent beliefs, one about permanent risk and one about transitory risk. The two cannot therefore be separately estimated from the perceived-variance series alone. I make two modelling choices to assign the dynamics and levels to these components.

First, I let the permanent-risk belief evolve over time, $\hat{\sigma}_{\psi,t}^2$, while holding the perceived transitory variance at a group-specific prior, $\sigma_{\theta,\text{perc},g}^2$. I made this decision since permanent shocks accumulate in the worker's earning power and therefore alter the entire future path of consumption, whereas transitory shocks disappear within the period and are largely absorbed by the household's buffer. The belief with the stronger saving implications should therefore be the one that responds to new information. A moving belief about transitory risk would add another latent state without much behavioural content in the buffer-stock economy. Moreover, the reading error in (7) is observationally equivalent to a transitory shock, which makes the transitory prior the natural place for uncertainty that the worker cannot resolve. This restriction comes at a cost since the perceived-variance series alone cannot test whether the transitory belief is actually static. Any belief-side misallocation between the permanent and transitory components is therefore absorbed

by the attribution structure of the model.

Second, I hold fixed the share of each squared surprise that the worker attributes to permanent earning power. Let λ_g denote this attribution share. The worker nets out labelled transitory pay before updating, so the remaining signal is cleaner than the raw pay change. A Bayesian worker would respond to this improvement in signal quality by increasing the share of the remaining surprise attributed to permanent risk. I instead assume that the worker applies the same attribution rule before and after netting.

This distinction is central to the mechanism. If the worker fully adjusted her attribution to the cleaner signal, labelling would improve information without creating a persistent bias in the level of perceived permanent risk. Under the constant-attribution rule, by contrast, the worker continues to treat the cleaner residual as though it contained the same mixture of permanent and transitory variation as the original pay change. Netting therefore reduces the amount of variation she attributes to permanent risk, generating a permanent-risk belief below the realised variance.

The constant attribution rule has a precedent in adaptive learning. For a random-walk state, a constant-gain updating rule is the standard perpetual-learning benchmark (Muth, 1960) and underlies much of the adaptive-learning literature (Evans and Honkapohja, 2001; Malmendier and Nagel, 2016). The restriction here is not the use of a constant gain itself, but the failure to adjust the attribution share when the information content of the signal changes. This is consistent with evidence that agents compress the informativeness of signals, updating too little on highly informative signals and too much on weak ones (Edwards, 1968; Augenblick et al., 2025). Noisy cognition can generate similar compression (Enke and Graeber, 2023).

4.3 The Belief Rule

The permanent-risk belief follows an exponentially weighted moving average of past squared wage changes:

$$\hat{\sigma}_{\psi,t}^2 = \underbrace{(1-\mu)}_{\text{persistence}} \hat{\sigma}_{\psi,t-1}^2 + \underbrace{\mu}_{\text{update weight}} \underbrace{\lambda}_{\text{attribution share}} (\Delta e_{t-1})^2. \quad (8)$$

where $\mu_g \in [0, 1]$ is the update speed and $\lambda_g \in [0, 1]$ is the share of each squared surprise that the worker attributes to permanent earning power. A higher μ_g makes the belief respond more strongly to recent news, while a higher λ_g raises the amount of each surprise attributed to the permanent component.

The recursion is a constant-gain learning rule. Its persistence is $1 - \mu_g$, so a shock to perceived risk decays geometrically over time. In squared shocks, it is analogous to an integrated GARCH(1,1) process. It can also be interpreted as a constant-gain approximation to a Kalman filter for learning about an evolving variance. Appendix D.1 derives the corresponding signal-extraction problem and relates the gain to the signal-to-noise ratio of the squared wage changes.

The survey asks about twelve-month earnings growth, so the worker maps her beliefs about the monthly permanent and transitory components into a twelve-month perceived variance:

$$\bar{\sigma}_{\text{perc},t}^2 = 12 \hat{\sigma}_{\psi,t}^2 + 2\sigma_{\theta,\text{perc}}^2 \quad (9)$$

The permanent component accumulates over twelve independent monthly shocks, while the transitory component enters at the two endpoints of the twelve-month change. This is the same aggregation used to construct the full-information benchmark in (5).

The learning rule also has a useful stationary interpretation. Taking unconditional expectations of (8) gives $\mathbb{E}[\hat{\sigma}_\psi^2] = \lambda \bar{S}_{m,g}$ so the permanent-risk belief is unbiased on average when $\lambda_g^* = \sigma_{\psi,g}^2 / \bar{S}_{m,g}$. Using the realised moments in Table 2, this full-information attribution share is 0.70, 0.57, and 0.49 for the bottom, middle, and top groups, respectively.

4.4 Legibility and Beliefs

I now connect how pay is delivered to the beliefs generated by the learning rule. The first result is that labelling reduces the share of an observed pay change that the worker attributes to permanent earning power. Labelled transitory pay contributes to the observed change but carries no information about permanent earning power. When the worker nets it out, the remaining change is smaller, but under the constant attribution rule she does not increase the share she assigns to permanent risk.

Proposition 1 (Attribution under partial labelling). *Suppose ψ and θ are jointly Gaussian and the labelled and unlabelled pieces are independent. Then the share of the full squared change a worker books as permanent is*

$$\lambda_g = \frac{\sigma_{\psi,g}^2 (\sigma_{\psi,g}^2 + 2(1 - \rho_g) \sigma_{\theta,g}^2)}{(\sigma_{\psi,g}^2 + 2\sigma_{\theta,g}^2)^2} = \lambda_g^* \frac{V_{u,g}}{\bar{S}_{m,g}}, \quad (10)$$

which is strictly decreasing in ρ_g and equals the full-information share λ_g^* at $\rho_g = 0$.

The proof is in Appendix B. The labelled component is part of the observed pay change but is removed before the worker forms her permanent-risk signal. As ρ_g rises, a larger part of the observed change is therefore uninformative about permanent earning power. Measured against the full wage change, the share attributed to permanent risk falls. This delivers a simple prediction: if pay becomes more legible as income rises, the attribution share should fall with income.

Proposition 2 (The permanent-belief discount). *Under the constant attribution maintained above, the belief (8) rests at*

$$\mathbb{E}[\hat{\sigma}_\psi^2] = \lambda_g^* V_{u,g} = \sigma_{\psi,g}^2 \frac{V_{u,g}}{\bar{S}_{m,g}} \leq \sigma_{\psi,g}^2, \quad (11)$$

with equality only at $\rho_g = 0$, and the discount $V_{u,g} / \bar{S}_{m,g}$ is strictly decreasing in ρ_g . The resting belief does not depend on the reading noise ω_g^2 .

The resting permanent-risk belief is therefore the realised permanent variance multiplied by the fraction of pay variance that remains after labelled transitory pay is removed. More legible pay produces a cleaner signal, but under the constant attribution rule the worker does not adjust her attribution share to account for this improvement. The resulting belief is consequently too low.

The absence of ω_g^2 from the permanent belief follows from a cancellation. Averaging the recursion

over a stationary state leaves the resting belief at the attribution times the mean of the signal. Reading noise inflates the mean of the signal, while the attribution is rescaled to that mean, so the noise cancels from the stationary permanent belief and survives only in the transitory prior.³

The second result concerns the transitory component. Netting labelled pay lowers the amount of transitory risk that remains in the worker's signal, while difficulty in reading the unlabelled remainder adds noise that is observationally equivalent to a transitory shock. The two forces therefore operate in opposite directions.

4.5 The Perceived-Risk Wedge

The previous results determine how legibility affects the two components of perceived risk. I now combine them to characterise the gap between perceived and realised twelve-month risk.

Define the perceived-risk wedge as the difference between the agent's perceived and realised second moments, using the weights from the twelve-month aggregation:

$$\Delta\sigma^2 = 12(\hat{\sigma}_{\psi}^2 - \sigma_{\psi,g}^2) + 2(\sigma_{\theta,\text{perc},g}^2 - \sigma_{\theta,g}^2). \quad (12)$$

Proposition 3 (Two-sided perceived risk). *The worker's transitory prior is her netting floor plus her reading noise,*

$$\sigma_{\theta,\text{perc},g}^2 = (1 - \rho_g) \sigma_{\theta,g}^2 + \omega_g^2, \quad \omega_g^2 \geq 0, \quad (13)$$

so the permanent gap $\hat{\sigma}_{\psi}^2 - \sigma_{\psi,g}^2 = \sigma_{\psi,g}^2 [V_{u,g}/\bar{S}_{m,g} - 1] \leq 0$ is one-signed, while the transitory gap $\sigma_{\theta,\text{perc},g}^2 - \sigma_{\theta,g}^2 = \omega_g^2 - \rho_g \sigma_{\theta,g}^2$ takes either sign. Total perceived risk can therefore sit on either side of the truth.

The permanent component is always under-perceived because labelling removes transitory variation from the signal without inducing the worker to increase the attribution share. The transitory component is different: labelling reduces the transitory variation that remains in the worker's signal, while reading noise adds variation that she cannot distinguish from a transitory shock. Hence, labelling pushes perceived transitory risk down, whereas reading noise pushes it up.

This two-sidedness is the key implication of the mechanism. A single directional source of belief error cannot generate a cross-section in which some groups over-perceive risk while others under-perceive it. Here the two forces operate in opposite directions: where pay is fragmented and difficult to read, reading noise can dominate and push perceived transitory risk above its realised counterpart; where pay is more legible, netting can dominate and push it below.

The crossing therefore occurs through the transitory component. The permanent component is always under-perceived, while the transitory component can lie on either side depending on the relative strength of legibility and reading noise. This allows total perceived risk to lie either above or below realised risk.

³The implemented reading is the third of three that share this resting mean. The agent updates on the squared change of her own noisy netted reading (7), whose mean is $V_{u,g} + 2\omega_g^2 = \sigma_{\psi,g}^2 + V_x$ with $V_x = 2\sigma_{\theta,\text{perc},g}^2$, and the attribution is the rescaled $\tilde{\lambda}_g = \sigma_{\psi,g}^2 V_{u,g}/[\bar{S}_{m,g}(\sigma_{\psi,g}^2 + V_x)]$, chosen so that the resting mean is unchanged (Appendix D.1, Table 11): the reading error inflates her signal and the rescaling offsets it, so it never reaches the resting permanent belief.

4.6 Prudence and Incidence

The perceived-risk wedge matters for behaviour through the curvature of the value function. Because the perceived and realised shocks have the same means, a second-order approximation isolates the effect of the difference in their variances:

$$\widetilde{\mathbb{E}} V - \mathbb{E} V \approx \frac{1}{2}(\hat{\sigma}_{\psi}^2 - \sigma_{\psi,g}^2)V_{\psi\psi} + \frac{1}{2}(\sigma_{\theta,\text{perc},g}^2 - \sigma_{\theta,g}^2)V_{\theta\theta}, \quad (14)$$

where $\widetilde{\mathbb{E}}$ and $\mathbb{E}$ denote expectations under the worker's perceived and the true distributions, respectively, and $V_{\psi\psi}$ and $V_{\theta\theta}$ are the corresponding curvatures evaluated at the mean. The equation identifies the channel through which a belief about risk enters the household problem, while the quantitative magnitude is to be determined by the model in Section 6.

The same channel can be seen from the Euler equation. A belief about future risk enters through the expected marginal utility of next-period consumption, $\widetilde{\mathbb{E}}[u'(c')]$. Expanding around mean next-period consumption gives

$$\widetilde{\mathbb{E}}[u'(c')] - \mathbb{E}[u'(c')] \approx \frac{1}{2}g_p\mathbb{E}\left[u'''(c')c_y^2 + u''(c')c_{yy}\right], \quad g_p \equiv \hat{\sigma}_{\psi}^2 - \sigma_{\psi,g}^2, \quad (15)$$

where c_y and c_{yy} are the first two derivatives of the consumption policy with respect to income. Under the CRRA preferences used below, the leading term is positive, so an agent who perceives less permanent risk than she faces under-saves for precaution.

The distinction between permanent and transitory risk is important for the incidence of the wedge. A transitory shock is largely absorbed within the period by the household's buffer, whereas a permanent shock changes the consumption path going forward. The variance error therefore matters much more when it concerns permanent rather than transitory risk. This is particularly important when the behavioural margin is precautionary saving: the permanent-risk wedge affects the desired buffer, while the transitory wedge has a comparatively small effect.

Proposition 4 (Incidence). *Let g_p denote the saving-relevant gap in (15), and let $a'(m, \hat{\sigma}_{\psi}^2)$ denote the saving policy. Then*

1. $\partial g_p / \partial \rho_g < 0$ and $\partial g_p / \partial \omega_g^2 = 0$: the saving-relevant distortion depends on legibility but not on reading noise;
2. only the permanent gap materially affects saving: a transitory belief error is buffered within the period and enters the consumption response at second order;
3. $\partial a' / \partial \hat{\sigma}_{\psi}^2 = 0$ at the borrowing constraint $a' = \underline{a}$ and is strictly positive only when the household has slack, whereas a level error reaches a constrained household one for one.

The first part follows from the cancellation in Proposition 2. Reading noise is recorded as transitory risk and therefore does not enter the saving-relevant permanent gap. Legibility, by contrast, lowers the permanent-risk belief and hence increases the gap between perceived and realised permanent risk.

The second part follows from how the two shocks affect consumption. A household with slack

can absorb a transitory draw using its buffer, so next-period consumption responds only weakly to it. A permanent shock instead shifts the entire future income path, making its effect on consumption much larger. Since a variance error enters through the squared consumption response, the transitory component has little saving relevance relative to the permanent component.

The third part concerns the role of the borrowing constraint. A constrained household already saves at its lower bound, so changing perceived risk does not change its actual saving. A level error is different: it changes the income that arrives and therefore reaches consumption one for one even when the constraint binds. If the constraint binds more often at low incomes, it further mutes the behavioural response of the bottom group.

Corollary 1 (The gradient in behaviour runs opposite to the gradient in beliefs). *Wherever legibility rises with income, the behavioural wedge rises with income. It is therefore largest for the group whose reported risk is lowest and smallest for the group whose reported risk is highest and most distorted.*

The corollary follows from the composition of the belief errors. The bottom group has the largest total distortion, but much of it comes from an inflated transitory prior that has little saving relevance. The top group reports less total risk, but its error is concentrated in the permanent component, which directly affects precautionary saving. The result is therefore that the gradient in behavioural distortions runs opposite to the gradient in reported risk.

To illustrate the incidence, consider a low-income household that is close to the borrowing constraint. Its perceived-risk error may be large, but if that error is primarily an overstatement of transitory risk, it has little effect on saving; and even an error in permanent risk has little effect while the constraint binds. A higher-income household may report less total risk, but if it has slack and under-perceives permanent risk, it responds by reducing its precautionary buffer. The largest belief distortion therefore need not generate the largest behavioural distortion: incidence depends both on which component is misperceived and on whether the household has room to adjust its saving.

4.7 Level Shocks versus Risk Shocks

The belief rule responds to realised wage shocks, but a level shock and a risk shock move perceived risk differently. A level shock changes the realised wage in a given period and so raises the squared surprise the worker updates on. A risk shock instead changes the variance of future wage growth, and so changes the distribution of every surprise that follows.

Under the belief updating rule (8), a level shock raises perceived permanent risk on impact by $\mu_g \lambda_g$ per unit of the squared change, and the effect decays geometrically, to $\mu_g \lambda_g (1 - \mu_g)^h$ after h months. A one-off wage shock therefore lifts perceived risk only temporarily.

A persistent rise in $\sigma_{\psi,g}^2$ works differently. Squared wage changes stay larger on average, so the worker keeps revising upward. Under the constant attribution she books only a fraction of the rise, so a true increase of $\Delta \sigma_{\psi}^2$ raises her perceived permanent variance by $\lambda_g \Delta \sigma_{\psi}^2$ alone, short of the truth.

This distinction matters for the dynamics of precautionary saving. A recession that produces

a temporary wage decline can raise perceived risk through the unusually large realised shock, but the effect fades as the worker receives subsequent observations. A persistent increase in income risk instead leaves households facing a sustained wedge between perceived and realised risk. The model therefore predicts a temporary belief response to level shocks but a persistent under-response to changes in the underlying risk process.

The mechanism can also vary over the business cycle. If downturns make pay less legible, for example because bonuses and commissions disappear while hours become more variable, the labelled share ρ_g falls and reading noise ω_g^2 rises. The resulting increase in perceived risk then reflects not only the lower level of income but also a change in how that income is delivered. A recession can therefore affect the perceived-risk wedge through two distinct channels: the realised wage shock changes beliefs directly, while changes in pay legibility alter the mapping from realised shocks into beliefs.

5 Estimation and Measurement

This section takes the belief rule to the data in two steps. I first estimate the learning rule within person by income group. I then measure the labelled share from itemised pay components and use it to test whether the mechanism can account for the estimated belief gradients.

The belief recursion (8) is written in the unobserved $\hat{\sigma}_\psi^2$. Combining it with the level identity (9) eliminates the latent belief and gives an AR(1) in the observable perceived variance:

$$\bar{\sigma}_{\text{perc},i,t}^2 = \underbrace{2\mu_g \sigma_{\theta,\text{perc},g}^2}_{\beta_{\text{cons}}} + \underbrace{(1 - \mu_g) \bar{\sigma}_{\text{perc},i,t-1}^2}_{\beta_{\text{lag}}} + \underbrace{12\mu_g \lambda_g (\Delta e_{i,t-1})^2}_{\beta_{\text{shock}}} + \alpha_i + u_{i,t}, \quad (16)$$

where α_i is a person fixed effect. The three coefficients map one-to-one to the structural parameters:

$$\mu_g = 1 - \hat{\beta}_{\text{lag}}, \quad \sigma_{\theta,\text{perc},g}^2 = \frac{\hat{\beta}_{\text{cons}}}{2\mu_g}, \quad \lambda_g = \frac{\hat{\beta}_{\text{shock}}}{12\mu_g}. \quad (17)$$

The person effects are normalised to have mean zero within each income group. Thus β_{lag} identifies the updating speed from within-person persistence, β_{cons} identifies the transitory prior from the group level, and β_{shock} identifies the response to realised wage news.

5.1 The Shock Regressor

The learning rule calls for the worker's own $(\Delta e_{i,t-1})^2$, which the SCE does not observe. I therefore use the SIPP cell mean of squared wage changes among workers in the same cell $c = (\text{income group} \times \text{Census division} \times \text{month})$. This captures wage news common to workers facing the same local labour-market conditions. Following Guerreiro (2023), I smooth the cell series with a three-month trailing moving average.

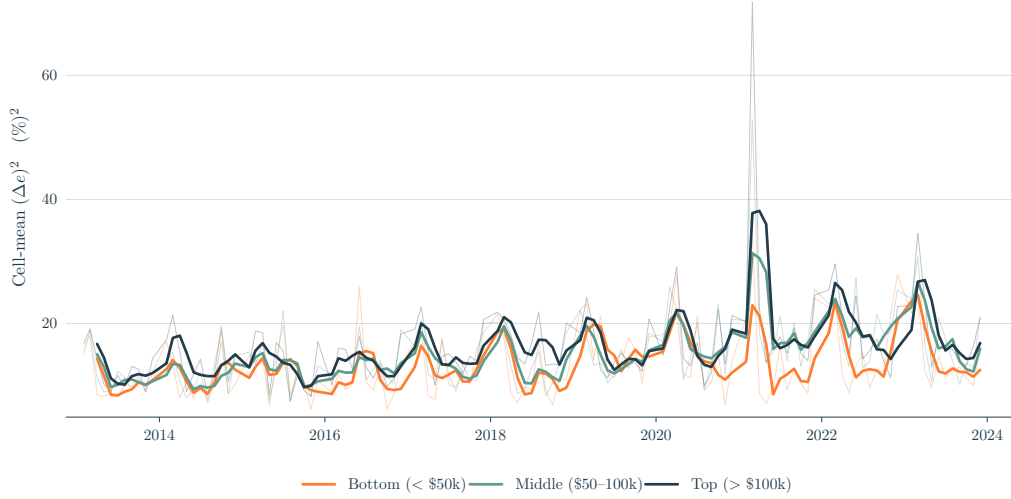


Figure 3: Cell-mean $(\Delta e_{i,t})^2$ by income group, national monthly, in $(\%)^2$. Thin lines: raw monthly cell means. Thick lines: the 3-month trailing moving average used as the regressor in (16).

Figure 3 shows that realised wage shocks are largest for the middle and top groups, consistent with their higher realised transitory variance and the concentration of discretionary pay at the top (Guvenen et al., 2021; Halvorsen et al., 2024). The bottom therefore does not receive larger shocks that could mechanically explain its stronger belief response. Its stronger response reflects how it updates to the shocks it receives.

The cell mean introduces measurement error because it captures only the component of a worker’s wage shock that is common within her cell. This attenuates β_{shock} , and therefore $\hat{\lambda}$, towards zero. The coefficient is consequently best interpreted as the response of perceived risk to common local wage news. Appendix D.3 estimates the group-specific loadings at 0.43, 0.56, and 0.58 for the bottom, middle, and top, respectively. Correcting for these loadings widens the cross-group gradient, so the estimated gradient is a lower bound on the underlying gradient.⁴

5.2 Estimates by Income Group

Joint GMM estimation of (16) by income group gives the estimates in Figure 4, which I set against the realised SIPP transitory variance. I treat the shock as endogenous, since it shares survey timing and aggregate news with the belief, and instrument it with its own lags and lags of perceived variance, as in the Blundell–Bond estimation of Table 1. The full estimates are reported in Table 5 (Appendix C).

⁴The loadings measure the covariance between individual and cell-level wage shocks relative to the variance of the cell-level shock.

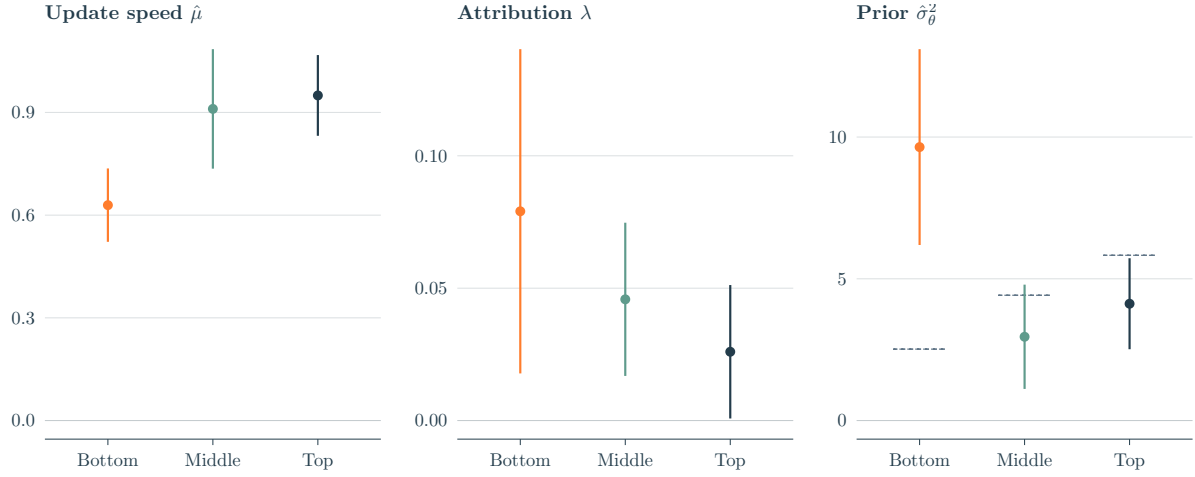


Figure 4: Headline estimates by income group: update speed μ_g , attribution λ_g , and transitory prior $\sigma_{\theta, \text{perc}, g}^2$, with 95% confidence intervals. Grey dashes in the prior panel mark realised transitory variance $\sigma_{\theta, g}^2$ (SIPP).

Update speed. The bottom updates significantly more slowly than the middle or top, matching the persistence ordering in Table 1. A given wage shock therefore remains in the bottom group’s perceived risk for longer.

Attribution. All three $\hat{\lambda}_g$ estimates are positive and statistically significant, and the point estimates decline monotonically with income. The bottom’s attribution share is about three times the top’s. The cross-group differences are not statistically significant, however, and part of the decline is consistent with full information, since λ_g^* itself falls with income, although at roughly half the estimated rate.

Transitory prior. Realised transitory variance rises with income, while the perceived prior is almost four times its realised counterpart at the bottom and lies below it in the middle and at the top. The cross-group gradient is statistically significant. The high level of perceived risk at the bottom therefore comes primarily from its transitory prior, whose source is examined in the second half of this section.

5.3 Robustness

I first add an occupation $\times$ year-month fixed effect to the shock construction. The persistence and attribution gradients remain, with the latter becoming sharper, while the prior gradient retains its direction but loses cross-group significance (Appendix C.1). I do not use this specification as the baseline because occupation is itself part of the mechanism: low-income workers sort into occupations with structurally more uncertain pay (Morduch and Schneider, 2017), so controlling for occupation removes part of the variation through which pay legibility may operate.

The estimates are also stable to changes in the instrument lags, trimming, and COVID treatment (Appendix C.2). The persistence gradient survives all variants except the tighter within-group trim, where its significance becomes marginal, and the attribution gradient remains monotone throughout. The bottom’s prior excess remains in every specification, ranging from 6.4 to 11.2

against a realised variance of 2.52, although its cross-group significance disappears under the COVID exclusion and the occupation control. I retain the COVID period because these years contain large realised wage movements that are part of the SIPP benchmark and provide identifying variation for the belief response.

Finally, I split the shock by sign, pooling income groups for power. Upside news receives roughly twice the weight of downside news. I treat this asymmetry as suggestive rather than as a central result, since pooling obscures which part of the income distribution drives it (Appendix C.3). If the asymmetry is actually present across all groups, I would interpret it as simple loss-aversion: workers appear to treat gains as more informative about persistent earning power than losses.

5.4 Measuring the Labelled Share

The estimates leave the three belief gradients organised but unexplained. The theory of Section 4 ties them to the labelled share ρ_g , which can be measured directly from the itemised pay components in the SIPP. I therefore use the same stayer panel that supplies the realised wage process to measure how much of each worker's pay movement arrives explicitly labelled as transitory.

The SIPP itemises the components of each paycheck, allowing me to measure the labelled share ρ_g directly on the same stayer panel used to estimate the realised income process. For each worker and month, I observe how much pay arrived as an identifiable component, such as a bonus or commission, and how much did not. I classify tips and overtime as unlabelled. The resulting share of labelled pay is 0.121 at the bottom, 0.178 in the middle, and 0.358 at the top. Appendix D.2 gives the construction and specification checks, which produce similar estimates across alternative definitions. These shares likely understate the extent of labelling because the denominator also contains changes in permanent earning power. Removing this component raises the shares to an upper bound of 0.176, 0.256, and 0.488.

These measured shares imply netting floors of 2.21, 3.63, and 3.74 (%)² from (13). The difference between each floor and the estimated transitory prior therefore identifies the reading noise. It is essentially zero for the middle and top groups, but 7.43 (%)² for the bottom group. The two sides of this calculation come from independent sources: the floors are measured from pay records, while the priors come from the SCE. The fact that the middle and top priors sit close to their netting floors supports the interpretation that labelled pay accounts for their lower perceived transitory risk, while the excess at the bottom points to the additional reading noise predicted by the mechanism.

5.5 Scoring the Estimates

I now ask whether the analytical results, evaluated at the measured labelled shares, can account for the estimated belief parameters. The filter predicts the updating gains, while Proposition 1 predicts the attribution shares.

Updating gains. The filter behind (8), evaluated at the measured shares, produces an implied updating gain for each income group. Greater legibility removes noise from the signal and should therefore allow the worker to update more quickly. The filter correctly predicts that the bottom updates most slowly, but understates the differences across groups: the implied gains are 0.76,

0.78, and 0.82 for the bottom, middle, and top, respectively, compared with estimated gains of 0.63, 0.91, and 0.95 (Appendix D.4). I therefore use the estimated gains in the quantitative model, while the filter provides a useful check on their ordering.

Attribution ratios. Proposition 1, evaluated at the measured shares, gives attribution shares of 0.677, 0.524, and 0.401, which decline with income as in Figure 4. The relevant comparison is across groups rather than in levels, since the cell-mean shock regressor attenuates the estimated response by a group-specific loading. The correction in Appendix D.3 increases the estimated cross-group gradient, so differential measurement error cannot account for the pattern. The predicted bottom-to-top and middle-to-top ratios are 1.69 and 1.31, compared with estimated ratios of 3.04 and 1.76, with log standard errors of 0.63 and 0.59. Both predictions therefore lie within one standard error of the estimates. The measured delivery of pay explains roughly half of the decline in attribution with income, leaving the remainder to either additional under-reaction not captured by labelling or sampling uncertainty.

5.6 The Belief Profile

The model of Section 4 maps the labelled share, the updating gain, and the transitory prior into belief profiles over income. Since income drifts continuously in the model, I interpolate these objects between the three estimated income groups. I then use the resulting profiles to construct the perceived-risk wedge and to parameterise the quantitative model.

I interpolate the labelled share and the updating gain in levels, and the prior in logs, using monotone interpolants through the three group estimates. Each object is held fixed beyond the observed income range. I also impose the netting floor from (13), censoring the prior below $(1 - \rho(y))\sigma_{\theta,g}^2(y)$. The attribution share and resting permanent-risk belief then follow from the closed-form expressions in Section 4. Appendix E.1 describes the interpolation and evaluates the resulting mapping at representative income levels. The prior retains its dip at the middle because the netting floors are themselves non-monotone.

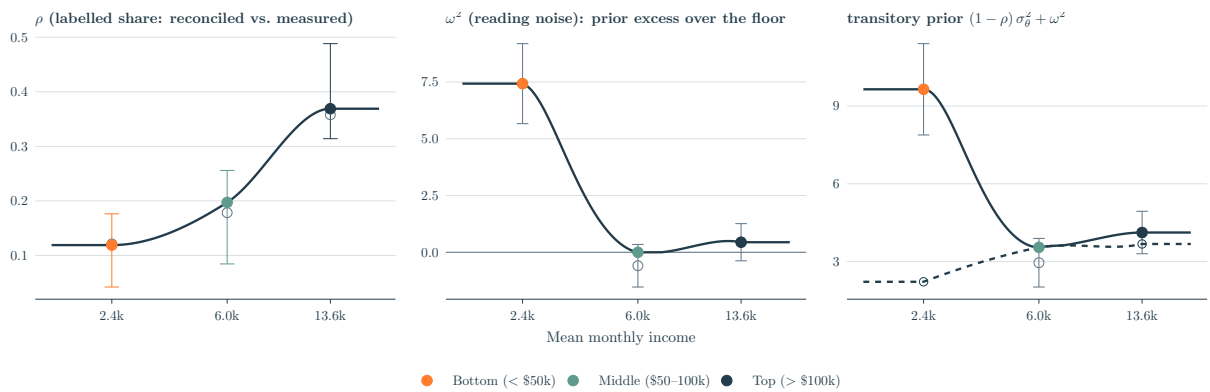


Figure 5: The labelled share ρ , the reading noise ω^2 , and the transitory prior they compose (13), by income. Left: the share used in the model (filled points and income interpolant) against the measured share (open points), with whiskers spanning the specification range in Appendix D.2. Middle: reading noise, measured as the prior's excess over its netting floor, with ± 1 standard-error bands in grey. Right: the carried prior, given by the netting floor plus reading noise, against the raw estimates and their ± 1 standard-error bands in grey. In $(\%)^2$ except for the dimensionless share.

The resulting profile determines the permanent and transitory components of perceived risk at each income level. Figure 6 uses these profiles to construct the perceived-risk wedge and compare it with the full-information benchmark. The figure confirms the qualitative pattern predicted by the theory: the permanent component is consistently under-perceived, while the transitory component crosses the benchmark for higher incomes, generating the predicted crossing in total perceived risk.



Figure 6: Perceived twelve-month risk against the FIRE benchmark, in closed form along the measured belief profile of Figure 5. Each panel draws the benchmark (solid) against its perceived counterpart (dashed), and the distance between the lines is the matching piece of the wedge (12): the permanent part $12\hat{\sigma}_{\psi,g}^2$ against $12\hat{\sigma}_{\psi}^2$ in (a), the transitory part $2\sigma_{\theta,g}^2$ against $2\sigma_{\theta,perc}^2$ in (b), and the total in (c). Points mark the three income groups. In $(\%)^2$.

6 The Belief Wedge in a Heterogeneous-Agent Economy

To trace what those perceptions imply for behaviour, I embed the estimated belief structure in a closed general-equilibrium incomplete-market economy and compare each income group under its estimated beliefs with an otherwise identical full-information benchmark. The gap between the two regimes, in precautionary wealth and the marginal propensity to consume, is the belief wedge. I set out the economy and its calibration first, then read the wedge in the stationary cross-section and in the closed economy, and close with the dynamics, where the belief is a state and the wedge moves with the two primitives it rests on.

6.1 Environment

Demographics and preferences. The economy is a monthly overlapping-generations life-cycle buffer-stock model following Wang (2023), closed with a single-asset technology. A worker enters at age 25, works for forty years, retires at 65 onto a pension replacing sixty percent of her permanent income after a one-fifth step-down in the permanent wage at the boundary, and dies with certainty at 85, with a small constant mortality risk before then. Each death is matched by a new entrant drawn across the income distribution, so the cross-section is stationary through the finiteness of life. Preferences are CRRA with $\gamma = 2$, and at the final age she values the assets she leaves directly rather than spending them to zero.

Income. Income follows the permanent-transitory process of Section 3 scaled by a deterministic

age-wage profile, $y_{i,t} = \exp(p_{i,t}) G_{\tau} \xi_{i,t}$. Here $p_{i,t} = p_{i,t-1} + \psi_{i,t}$ is the permanent random walk and $\xi_{i,t}$ the transitory state, with their variances taken from Table 2, while G_{τ} is the hump-shaped age profile Wang (2023) uses to separate predictable life-cycle earnings growth from risk. When employed the worker draws $\xi = \exp(\theta_{i,t})$; when unemployed she receives a benefit replacing half her permanent income, and in retirement she receives the pension.

Mobility. The permanent innovation arrives while she is employed, consistent with the same-job stayer panel its variance is measured on, and it is the shock she learns about in Section 4. Accumulated over a career, the innovations carry her across the income distribution: a run of positive permanent raises lifts a worker out of the bottom group without any change of job, and the deterministic profile G_{τ} adds a predictable component to the same drift. A current-income tercile therefore mixes types, and the results return to the distinction. The anchors themselves are household-income groups, which the model reads as positions in individual earning power. Appendix C.6 checks the realised side of the mapping by re-sorting on own earnings.

Unemployment. Caplin et al. (2026) find that the possibility of quitting or being fired is one of the key drivers of subjective earnings risk, so the transition margin cannot be left out of a model of perceived risk. The SCE question conditions it away by fixing the job, which is what makes the wage-risk measurement clean, so following Wang (2023) the budget restores it through a two-state employment block, with rates taken from the SCE’s own perceived job-loss and job-finding probabilities. A second finding of Caplin et al. (2026), that average subjective separation probabilities and expected durations track administrative records even by age, is what allows perceptions to supply them.⁵ The rates come to monthly separation 0.015/0.011/0.011 and job-finding 0.23/0.27/0.29 from bottom to top, implying a steady-state unemployment rate of 6.2/4.0/3.7 percent, higher and more persistent at the bottom and a little below the seven-percent flow benchmark of Shimer (2005). The transition margin is therefore perceived accurately where wage risk is not. Because its gradient is common to both regimes, it shapes the wealth distribution, the hand-to-mouth share, and the constraint incidence the wedge acts through without being a source of the wedge, as Wang (2023) emphasises. Beliefs update only while employed and freeze during unemployment.

Beliefs. The labelled share, the update gain, and the perceived transitory prior are functions of current income, $\rho(y)$, $\mu(y)$ and $\sigma_{\theta, \text{perc}}^2(y)$, interpolated through the tercile anchors of Section 5, and a worker slides along them as her income drifts.

Normalisation. Following Carroll (2006) I normalise by permanent income: the state is cash-on-hand per unit of permanent income, the per-period draw is the transitory $\exp(\theta)$, and permanent shocks enter through the growth factor $G_{\tau} \exp(\psi)$. The problem is scale-invariant, so I report each group’s mean monthly SCE income (\$2,415 / \$6,014 / \$13,604) only to fix magnitudes.

Technology and market clearing. In general equilibrium I close the economy with a constant-returns technology, $Y = ZK^{\alpha}N^{1-\alpha}$, capital share $\alpha = 0.36$ and monthly depreciation from an annual 2.5

⁵For each employed respondent I take the SCE’s reported percent chance of losing the main job over the next twelve months and, conditional on losing it, of finding a new job within three months, average each within income group with survey weights, and convert a horizon- H probability P to a monthly hazard $-\log(1 - P)/H$ (with $H = 12$ for separation and $H = 3$ for finding), which is the conversion Wang (2023) applies to the same SCE questions. The tercile-mean twelve-month job-loss probabilities are 16.9/12.7/12.2 percent and the three-month finding probabilities 50.2/56.0/57.6 percent.

percent, with the single asset carrying both the household buffer and the return. The unemployment benefit and the pension are financed by flat balanced-budget taxes on labour income, so disposable income is net of a payroll and a social-security rate. The capital market clears against household asset supply, and I pin the productivity normaliser Z once from the FIRE economy at a target rate of three percent, holding it fixed across regimes so the FIRE-versus-belief rate gap is the general-equilibrium wedge.

Two regimes. I solve the model twice. Under the *belief* regime the agent perceives wage risk through the estimated rule and her belief evolves; under *FIRE* she knows the true variances. Everything else is shared, so any gap is the belief wedge. FIRE here means correct wage-risk beliefs.

6.2 The Household Problem

$$V_\tau(m, p, s, \hat{\sigma}_\psi^2) = \max_{0 \leq c \leq m} \left\{ u(c) + \beta (1 - D) \tilde{\mathbb{E}} \left[\Gamma'^{1-\gamma} V_{\tau+1}(m', p', s', \hat{\sigma}_{\psi'}^2) \right] \right\},$$

$$m' = \frac{R}{\Gamma'} (m - c) + \xi', \quad \Gamma' = G_\tau \exp(\psi').$$

where m is normalised cash-on-hand, τ age, p the income node, s employment status, D monthly mortality, $R = 1 + r$ the gross return, and $\tilde{\mathbb{E}}$ the expectation under the agent's *perceived* process. The terminal age carries the bequest value in place of a continuation value. The budget uses *realised* income while expectations are taken under the perceived process, so beliefs act only through the saving motive (Section 4).

The belief follows the constant-gain rule of Section 4, $\hat{\sigma}_{\psi,t}^2 = (1 - \mu(y)) \hat{\sigma}_{\psi,t-1}^2 + \mu(y) \tilde{\lambda}(y) (\Delta \tilde{y}_t)^2$, on the agent's own netted reading $\tilde{y}$ of (7), with the attribution rescaled to the mean of that reading so that the belief rests at the value (11) implies (the rescaling is in Appendix D.1). Table 3 collects each input. The steady-state comparison uses this resting value, which adds no state dimension. The dynamic state is carried explicitly for the risk-shock response.

Stationary equilibrium. For each regime a stationary equilibrium is a family of age-indexed value functions, policies $c_\tau(m, p, s, \hat{\sigma}_\psi^2)$, and an invariant distribution Φ such that the policies solve the household problem, Φ is invariant under them, the capital market clears at the equilibrium interest rate r , and the government budget balances. The invariance runs over the income process, the demographics, the employment transitions, and the belief law of motion, which is degenerate at the truth under FIRE. I read moments of Φ off a simulated panel of $N = 30,000$ households over 1,200 months (720 burn-in), with the belief and FIRE branches driven by a single common shock path. Each worker then sits in the same income tercile under both regimes, and the belief-minus-FIRE gap is a within-worker difference free of sampling noise.

6.3 Calibration

The belief block is carried as Section 5 delivers it, the gain and the transitory prior estimated, the labelled share pinned inside its measured band, and the attribution and the resting permanent belief implied in closed form by (10) and (11), within half a point of their measured-share values. The calibration fits a single remaining parameter: a discount factor common across agents, matched to average liquid wealth. Everything else is assigned or carried as measured, and Table 3 collects the values. The model is block-recursive: beliefs evolve on the realised income process

and never on assets or consumption, so the belief block is measured and estimated independently of preferences.

Parameter	Symbol	Value (Bot / Mid / Top)	Source or target
<i>A. Preferences and assets (common)</i>			
Risk aversion	γ'	2	Assigned (standard)
Annual interest rate	r	2%	Fixed for PE; clears near 3% in GE
Borrowing limit	$\underline{a}$	0	Assigned (standard)
Discount factor	β	0.917	Calibrated to mean liquid wealth (≈ 1.8 mo)
<i>B. Employment block (heterogeneous by income; common across regimes)</i>			
Separation rate	s_g	0.015/0.011/0.011/mo	SCE perceived, monthly hazard
Job-finding rate	f_g	0.23/0.27/0.29/mo	SCE perceived, monthly hazard
Replacement rate	b	0.5	Standard convention
Pension replacement	ς	0.6	As in Wang (2023)
<i>C. Realised income process by group, (%)²</i>			
Permanent variance	$\sigma_{\psi,g}^2$	11.9/11.6/11.2	SIPP (Table 2)
Transitory variance	$\sigma_{\theta,g}^2$	2.5/4.4/5.8	SIPP (Table 2)
Mean monthly income	–	\$2,415/6,014/13,604	SCE (scale anchor only)
<i>D. Beliefs by group: the labelled share pinned, the rest implied or carried</i>			
Labelled share	ρ_g	0.119/0.197/0.369	Pinned in measured band (Section 5)
Update gain	μ_g	0.629/0.910/0.950	GMM estimate, carried
Attribution	λ_g	0.677/0.519/0.398	Implied from ρ_g , eq. (10)
Transitory prior	$\sigma_{\theta,\text{perc},g}^2$	9.65/3.55/4.12	GMM estimate, floor-censored (Mid binds)
Resting permanent belief	$\hat{\sigma}_{\psi}^2$	11.47/10.61/9.11	Implied, eq. (11)

Table 3: Model calibration. The employment block is heterogeneous by income but common to both regimes; under FIRE the belief block of panel D is held at the truth.

Discount factor. I set the discount factor so that the FIRE economy, at the fixed rate of 2 percent annual, matches the average liquid wealth in the data, about 1.8 months of permanent income, which gives an annual β of 0.917.⁶ A single calibrated β pins the *level* of wealth and of the marginal propensity while leaving the cross-section of both an outcome of risk. The alternative of a consensus discount factor, as in Wang (2023), lets the economy over-accumulate, and an income-varying $\beta(y)$ would build the cross-sectional gradient into preferences.⁷

6.4 The Cross-Sectional Wedge

Implied perceived risk. Table 12 (Appendix F) sets the stationary perceived variance against FIRE in the same SIPP units, with prices fixed at the calibrated rate, so the comparison is internal to the model, free of the level gap between the two surveys, and the gap between the regimes is the belief channel alone. The model's perceived variances accordingly sit seven to thirteen times above the

⁶Targets from the SCE Household Finance module, using the liquid-wealth concept of Kaplan et al. (2014): median liquid wealth about 0.5/2.0/3.0 months across the three groups, averaging near 1.8.

⁷Calibrating on the FIRE regime is a normalisation: under the measured beliefs average wealth falls short of the FIRE target by a few percent, so recalibrating on the belief regime would leave the wedge essentially unchanged. The resulting β sits below the 0.95–0.96 of the life-cycle literature (Gourinchas and Parker, 2002), which is the price of holding a one-asset economy to 1.8 months of *liquid* wealth rather than to total net worth. I solve by the endogenous grid method (Carroll, 2006).

reported levels of Section 3, the convention stated in Section 5. The reported shortfall lives in the compounded permanent term and is read as an upper bound, the intercept crosses the surveys unrescaled, and carrying the reported level itself would hand every group an under-perception so large that no group difference could matter. Perceived twelve-month risk falls with income, from 153 through 139 to 124 against a FIRE benchmark flat near 148, a ratio of 1.03/0.94/0.84. The bottom perceives about three percent more total risk than it faces, its inflated transitory prior outweighing a permanent discount of about four percent, while the middle and the top perceive six and sixteen percent less: the labelling discount deepens with income faster than the prior falls, as Section 4 derives.

Figure 7 compares the belief regime with an otherwise identical FIRE benchmark at the calibrated interest rate, and Table 13 (Appendix F) reports the numbers. With the discount factor matched to average liquid wealth, the FIRE economy holds 2.3/1.7/1.5 months of permanent income, with twelve-month MPCs of 0.47/0.41/0.38 and hand-to-mouth shares of 43/41/42 percent; the propensities are in the data's range and the shares a little above it. Mean wealth in months falls mildly with income where the median in the data rises, and the miss survives the workers-only split of Table 13.⁸ The belief wedge is computed with this calibration held fixed.

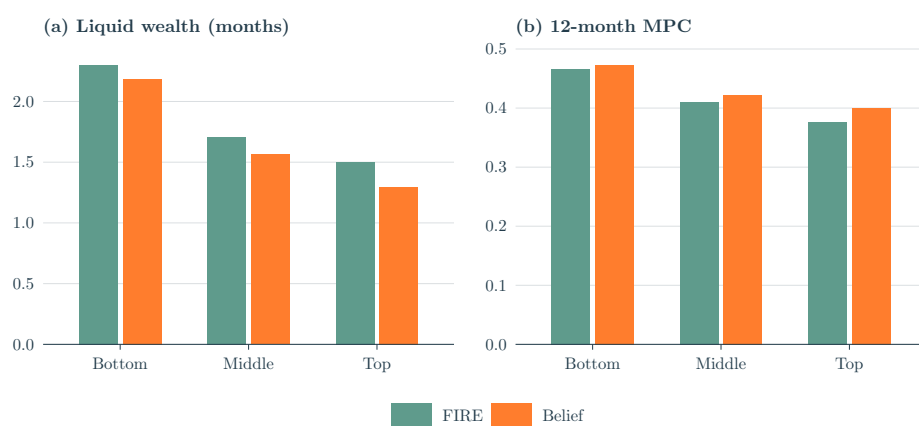


Figure 7: Mean liquid wealth (months of permanent income) and the 12-month MPC by income group, FIRE versus the belief regime (Table 13).

The wedge in wealth rises monotonically with income: the belief economy holds 5.1, 8.3, and 13.8 percent less than FIRE from bottom to top under the reconciled shares, and 7.0, 10.9, and 17.6 percent less at the detrended upper edge of the measurement band (Appendix D.2). The wedge in the marginal propensity to consume concentrates at the top, +0.7, +1.3, and +2.4 points, and +3.1 points at the upper edge. Both profiles track the permanent-belief gap, smallest at the bottom and largest at the top, and the wedge per unit of belief gap is nearly flat across groups, so liquidity adds little.

The wedge is robust to the rate it is read at: recalibrating the discount factor to the same liquid-wealth target at the general-equilibrium rate of three percent leaves it at 5.1, 8.3, and 13.6 percent,

⁸The direction is the one-asset buffer at work: wealth in months is a buffer against risk, and the risk that matters most, persistent unemployment, falls with income, and the rising liquid-wealth profile in the data reflects the life-cycle, illiquid, and bequest saving that a single asset cannot separate from the buffer. Income-varying patience would reproduce that profile; the common discount factor forgoes it to keep the cross-section an outcome of risk.

so the wealth calibration pins the cross-section at either rate. Holding the discount factor fixed instead, the rate alone would widen it to 6.8, 10.5, and 16.4 percent. A decomposition attributes the wedge entirely to the permanent belief. Solving the model with the permanent discount alone, the transitory prior held at the truth, reproduces the full wedge on both margins, while the complementary regime, which carries the prior and holds the permanent belief at the truth, moves neither wealth nor the marginal propensity measurably. The prior is behaviourally inert, and the permanent-belief discount accounts for the whole wedge.

The bottom is least moved on either margin, +0.7 points on the MPC and 5.1 percent on wealth, much as Corollary 1 anticipates: a bottom-income type labels little, so its permanent belief is discounted by under four percent and carries the least saving-relevant distortion. Composition does part of the rest. Just over half of the bottom current-income tercile is retirees, against roughly a quarter and a tenth above, and among its workers the bottom is the least constrained of the three groups, with hand-to-mouth shares of 27/38/38 percent from bottom to top. The smaller wedge is therefore a fact about beliefs and composition, with the borrowing constraint playing little part. Restricting to workers and re-terciling among them sharpens the shape mildly, to $-5.8/-9.7/-15.1$ percent on wealth and $+0.9/+1.5/+2.7$ points on the marginal propensity.

The middle is moderately distorted on both margins, -8.3 percent on wealth and $+1.3$ points on the MPC: unconstrained and labelling more of its transitory pay, it under-saves. The top responds most on both, holding 13.8 percent less wealth and consuming over two points more of a windfall over the year: it labels its transitory pay away most completely, under-perceives the permanent risk that drives saving most, and so holds the most-discounted permanent belief and the largest wedge. The permanent belief lies below the truth at all times, so the wedge is a property of the stationary state, in place before any shock arrives.

The size of the wedge turns on the reading of the updating rule, with sampling error second order. Under the Bayesian reading, whose agent raises her attribution as netting cleans her signal, the permanent belief is unbiased and the wedge is zero by construction, so the quantitative results stand or fall with the constant attribution and the evidence of Appendix D.4 that favours it. Under the constant attribution at the measured shares, the wedge is the baseline of 5.1, 8.3, and 13.8 percent of wealth; under the belief-anchored variant of Appendix E.2, which takes the estimated attribution responses at face value, it roughly doubles. The shape, rising with income and concentrated at the top, is common to both, and I read the baseline as the estimate and the variant as the outer edge, because only the baseline rests on the measured delivery of pay.

6.5 The Closed Economy

Closing the economy lets the saving wedge move the interest rate. At the calibrated discount factor the FIRE economy clears at the three-percent target by construction, with the productivity normaliser held fixed across regimes as the setup requires. The belief economy, which under-perceives permanent risk, supplies about five percent less aggregate capital and clears at a rate 0.21 percentage points higher, 3.2 against 3.0 percent (Table 14, Appendix F). The response corroborates Wang (2023) in general equilibrium: lower perceived risk lowers precautionary supply and raises the equilibrium return. Under the measured shares the movement is modest, however, and the wedge shows in the cross-section more than in the aggregates. The magnitude is the one-

asset response, in which a single liquid asset carries both the buffer and the return. A credible quantification of how the wedge moves marginal propensities and the transmission of aggregate shocks needs a two-asset closure that separates liquid from illiquid wealth (Kaplan and Violante, 2022).

6.6 Dynamics

The belief is a state, so the wedge moves as the belief moves. It responds to the two primitives the belief rests on: the legibility of pay, which is the share ρ a worker can label and net out, and the permanent risk σ_ψ^2 the pay carries. I trace a change in each through the belief channel with prices held fixed, setting the dynamic-belief economy of this section against an otherwise identical full-information economy on the same realised path. The delivery change is the mechanism's own experiment and comes first, the risk change is the aggregate experiment of the literature read through the mechanism, and a recession, which changes neither primitive, closes the section as a check.



Figure 8: Response to a legibility reform that raises the labelled share of the bottom and the middle to the top's level of $\rho = 0.37$, belief regime, reform minus no-reform, by income group. Perceived risk is the twelve-month perceived variance in $(\%)^2$, and consumption and liquid assets are in percent of the no-reform path. Paths average twelve simulated cohorts, and the perceived-risk panel is a three-month moving average of the belief-state mean.

A legibility shock. The experiment the mechanism most directly invites changes how legibly pay is delivered and leaves the risk itself untouched. Let a reform raise the labelled share ρ of a low-paid worker to the level a salaried worker already holds, so that variable shift counts and cash supplements arrive as a single itemised figure while the true income process is unchanged. The full-information economy then does not move, and the entire response is the belief channel, shown in Figure 8. As the cleaner signal arrives, the worker nets more of her pay before she updates and reads what remains as less telling about her permanent earning power, and her perceived permanent risk falls within a quarter, sharply for the newly legible bottom and middle and barely for the top. She then saves less against a risk she now under-perceives: consumption rises on impact and the precautionary buffer erodes over one to two years, by 2.9 and 3.0 percent for the bottom and the middle against 1.1 for the top, whose smaller move arrives through the treated

workers who drift into it. The belief adjusts within months and the buffer over years, and the figure separates the two speeds.

The incidence follows which workers the reform reaches. Formalising low-paid work moves the bottom and the middle towards the under-perception the top already carries, so the response lands on them. A broad rise in legibility across all pay would instead fall hardest on the top, where the labelled part scales with the largest transitory variance. Either way the direction is the same: making pay more legible lowers precautionary saving, because a cleaner permanent signal, read under an attribution that does not re-scale to it, registers as less risk. The response is accordingly conditional on the constant attribution, since the Bayesian of Section 4, re-scaling as netting cleans her signal, would leave her saving unmoved. The value of the exercise is the design rather than the magnitude: the steady-state wedge lives in a belief level no survey observes, while the response to a delivery reform is a behaviour a payroll change with saving data could confirm or refute.



Figure 9: Response to a persistent 40% rise in the true permanent wage variance, belief regime minus FIRE, by income group. Perceived risk is the gap in the rise in twelve-month variance in $(\%)^2$, and consumption and liquid assets are in percent of the no-shock baseline. Paths average twenty-four simulated cohorts, and the perceived-risk panel is a three-month moving average of the belief-state mean.

A risk shock. The mechanism also governs how the distribution absorbs a rise in the risk itself. Let the true permanent variance rise persistently, by forty percent, borrowing the magnitude from the literature (Bayer et al., 2019). Full information recognises the higher risk at once and rebuilds its buffer, holding fifteen to nineteen percent more liquid wealth within two years. The belief regime books only a fraction of the now-larger squared surprises as permanent, 0.37/0.57/0.47 across the groups, smallest at the bottom, where the noisy reading damps it, and largest at the middle, where neither the reading noise nor the netting is strong. Its perceived permanent risk therefore rises far less, and it accumulates 14.7/18.1/18.7 percent less additional liquid wealth than full information over the two years, each regime read against its own no-shock path (Figure 9). The shortfall is the same under-reaction the legibility reform exploits, now against a moving truth rather than a moving signal, and the precautionary response to rising risk is muted most at the top, where legibility is highest.

A recession, by contrast, changes neither primitive. Modelled as a one-off three-percent fall in the level of income, it delivers a single surprise that lifts the belief only transiently, as Section 4 anticipates, and the two regimes leave the episode with the same buffer. The wedge lives in the primitives the belief rests on, how pay is delivered and how risky it is, rather than in the business cycle.

7 Conclusion

This paper set the risk workers perceive against the risk their wages carry, on the same jobs, and found that the assumption equating the two fails systematically across the income distribution. Workers at the bottom perceive about double the wage risk of workers at the top, hold that belief more persistently, and revise it more strongly after a realised wage shock, while the risk itself, if anything, rises mildly with income. A constant-gain rule estimated within person organises all three gradients and locates the deviation on both sides of the truth: the bottom reports transitory risk far above its realised counterpart, the middle and the top under-perceive their permanent risk, and reported totals lie below the full-information benchmark in every group.

I propose that the gradient reflects how pay is delivered. The share of pay movement that arrives labelled on the paycheck rises from about an eighth at the bottom to over a third at the top, and a worker with legible pay nets the labelled part out before she updates: her transitory prior falls to the unlabelled remainder, her belief moves faster, and she books less of each surprise as permanent. Measured from the itemised pay components of the same records, the share predicts where the middle and top priors settle, to within a standard error, the ordering of belief persistence, and about half of the estimated gradient in the response to news; the bottom's prior stands four standard errors above the level its share implies, and the mechanism attributes the excess to the unresolved reading of fragmented pay.

In the life-cycle economy the deviation becomes a tilt in the cross-section: a calibration that equates perceived with realised risk overstates precautionary wealth and understates the propensity to consume by amounts graded in income. The error is negligible at the bottom, whose permanent belief is nearly unbiased and whose inflated transitory prior the buffer absorbs, and largest at the top, which under the measured beliefs holds fourteen percent less liquid wealth and consumes over two points more of a windfall within the year than under full information. The spread of propensities between the bottom and the top narrows by about a sixth, so an exercise whose shock or policy falls on higher incomes under-predicts the consumption response of the households the calibration assigns the lowest propensities; the aggregates move little, the capital stock by five percent and the interest rate by about twenty basis points, so the correction is distributional. Where the question involves shocks to risk itself, the calibration also overstates the precautionary response to them, since agents book only part of a rise in risk as permanent and rebuild fifteen to nineteen percent less buffer than full information within two years.

The recommendation for the practitioner is then direct: keep the realised process in the budget, and hand the agent a permanent variance discounted below it by an income-graded factor dependent on their position in the income distribution. The transitory side stays at the truth, and the discount adds no state variable, so the whole correction enters a standard calibration as a change

of parameters.

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A 12-Month FIRE Variance

Rolling $e_t = p_t + \theta_t$ forward H months, the permanent piece accumulates one fresh innovation per month, $\text{Var}(p_{t+H} - p_t) = H\sigma_{\psi,g}^2$, while the iid transitory piece contributes only its endpoints, $\text{Var}(\theta_{t+H} - \theta_t) = 2\sigma_{\theta,g}^2$. The shocks are independent, so $\text{Var}(e_{t+H} - e_t) = H\sigma_{\psi,g}^2 + 2\sigma_{\theta,g}^2$, which at $H = 12$ is (5).

Group	FIRE benchmark ($12\sigma_{\psi,g}^2 + 2\sigma_{\theta,g}^2$)	SCE observed $\bar{\sigma}_{\text{perc}}^2$	ratio
Bot	147.7	20.6	7.2×
Mid	148.0	11.3	13.1×
Top	146.3	11.4	12.8×

Table 4: FIRE-implied 12-month variance versus SCE-reported perceived variance, in (%)².

Appendix C.7 recomputes the benchmark under the same-hours restriction, where it falls to 79, 85, and 98 (%)² and the ratios to 3.8×, 7.5×, and 8.6×.

B Closed Form for λ_g under Partial Labelling

This appendix derives the closed form (10): the attribution on the full pay change is the unbiased attribution on the unlabelled residual, attenuated by the labelled pay that rides in the change. Split the transitory shock $\theta_t = \theta_t^l + \theta_t^u$ into independent pieces with variances $\rho_g\sigma_{\theta,g}^2$ and $(1 - \rho_g)\sigma_{\theta,g}^2$. Then $\Delta e_t = u_t + l_t$, with unlabelled residual $u_t = \psi_t + \theta_t^u - \theta_{t-1}^u$ and labelled component $l_t = \theta_t^l - \theta_{t-1}^l$, so $V_{u,g} = \sigma_{\psi,g}^2 + 2(1 - \rho_g)\sigma_{\theta,g}^2$, $V_{l,g} = 2\rho_g\sigma_{\theta,g}^2$, and $V_{u,g} + V_{l,g} = \sigma_{\psi,g}^2 + 2\sigma_{\theta,g}^2 = \bar{S}_{m,g}$, which is the variance of the full monthly change.

The rule of Section 4 updates on the full observed change $(u + l)^2$, so the attribution recovered by (16) is the unbiased residual attribution $\sigma_{\psi,g}^2/V_{u,g}$ attenuated by the projection of u^2 on $(u + l)^2$, $\lambda_g = (\sigma_{\psi,g}^2/V_{u,g}) \text{Cov}(u^2, (u + l)^2)/\text{Var}((u + l)^2)$. Under joint Gaussianity with $u \perp l$, $\text{Cov}(u^2, (u + l)^2) = 2V_{u,g}^2$ and $\text{Var}((u + l)^2) = 2(V_{u,g} + V_{l,g})^2$, so

$$\lambda_g = \frac{\sigma_{\psi,g}^2 V_{u,g}}{(V_{u,g} + V_{l,g})^2} = \frac{\sigma_{\psi,g}^2 (\sigma_{\psi,g}^2 + 2(1 - \rho_g)\sigma_{\theta,g}^2)}{(\sigma_{\psi,g}^2 + 2\sigma_{\theta,g}^2)^2},$$

which is (10).

This closed form is one of three readings of the estimating equation, and the scorecard's ratio comparison does not turn on the choice among them. Under the netting reading, in which the agent nets first and updates on u_t^2 at the fixed attribution $\lambda_g^* = \sigma_{\psi,g}^2/\bar{S}_{m,g}$, the coefficient in (16) is the projection $\lambda_g^* (V_{u,g}/\bar{S}_{m,g})^2 = \sigma_{\psi,g}^2 V_{u,g}^2/\bar{S}_{m,g}^3$. The predicted bottom-to-top and middle-to-top ratios become 1.99 and 1.48 in place of 1.69 and 1.31, and both pairs lie within one standard error of the estimated ratios (Appendix D.3). The implemented form, updating on the agent's own noisy netted reading with the rescaled attribution $\tilde{\lambda}$ (Appendix D.1), shares its resting mean with both, so the readings differ in the coefficient and not in the resting belief.

The Gaussian fourth-moment identities are checked directly on the component panel of Ap-

pendix D.2: replacing them with the sample projection $\text{Cov}(u^2, (u + l)^2)/\text{Var}((u + l)^2)$ moves the predicted attribution ratios from 1.69 and 1.31 to 1.54 and 1.13 under the baseline trim, still about one standard error from the estimated ratios, while the untrimmed projection is dominated by the labelled bonus tail documented there.

C Belief Estimates and Robustness

Table 5 reports the headline joint-GMM estimates of (16) plotted in Figure 4, C.1–C.6 check their robustness, C.7 constructs the same-hours counterpart of the realised process, and C.8 measures the twelve-month benchmark directly.

Group	$\sigma_{\theta,g}^2$ (SIPP truth)	$\sigma_{\theta,\text{perc},g}^2$ (GMM)	λ_g (GMM)	μ_g (GMM)	Hansen J p	N
Bot	2.52	9.65 (1.76)	+ 0.079 (0.031)	0.629 (0.055)	0.23	1,531
Mid	4.42	2.96 (0.94)	+0.046 (0.015)	0.910 (0.089)	0.23	7,501
Top	5.83	4.12 (0.82)	+0.026 (0.013)	0.950 (0.060)	0.23	11,331
Wald ($B = T$)	—	$p = \mathbf{0.004}$	$p = 0.12$	$p = \mathbf{0.0001}$	—	—

Table 5: Joint Blundell-Bond system GMM on (16), by income group (the estimates of Figure 4). Two-way clustered SEs (respondent, quarter) in parentheses. Instruments: $\bar{\sigma}_{\text{perc}}^2$ at lags 3–4, the shock at lags 1–3, plus BB level moments, collapsed. N counts observations, and respondent counts are 269/1,050/1,452. The Hansen J is a single joint-system statistic repeated across rows. Bold p -values mark Wald rejections of cross-group equality at 5%, and bold estimates mark the bottom’s prior and attribution (Section 5).

C.1 Occupation $\times$ Time Fixed Effect

Table 6 adds an occupation $\times$ year-month FE to the shock construction: the persistence and attribution gradients survive, attribution sharpens, and the prior gradient keeps its sign but loses significance.

Group	Canonical			Occupation $\times$ time FE		
	$\hat{\mu}_g$	$\hat{\lambda}_g$	$\hat{\sigma}_{\theta,\text{perc},g}^2$	$\hat{\mu}_g$	$\hat{\lambda}_g$	$\hat{\sigma}_{\theta,\text{perc},g}^2$
Bot	0.629 (0.055)	+ 0.079 (0.031)	9.65 (1.76)	0.649 (0.061)	+ 0.088 (0.039)	7.30 (2.66)
Mid	0.910 (0.089)	+0.046 (0.015)	2.96 (0.94)	0.923 (0.090)	+0.037 (0.010)	2.62 (0.98)
Top	0.950 (0.060)	+0.026 (0.013)	4.12 (0.82)	0.959 (0.060)	+0.017 (0.008)	4.19 (0.73)
Wald μ (T–B)		$p = \mathbf{0.0001}$			$p = \mathbf{0.0003}$	
Wald λ (B–T)		$p = 0.116$			$p = 0.068$	
Wald σ^2 (B–T)		$p = \mathbf{0.004}$			$p = 0.260$	

Table 6: Canonical versus occupation $\times$ year-month FE in the wage-shock construction. Standard errors in parentheses.

C.2 Lag Depth, Trim, COVID, and the Bottom’s Trend

Specification variants. Table 7 varies the lags, trim, and COVID handling one at a time: the Hansen J never rejects, attribution stays monotone in every row, the persistence Wald survives all but the tighter trim, the prior Wald all but the COVID exclusion.

Variant	$\hat{\mu}_g$			$\hat{\lambda}_g$			$\hat{\sigma}_{\theta, \text{perc}, g}^2$			Wald p (B vs T)		
	B	M	T	B	M	T	B	M	T	μ	λ	σ^2
Canonical	0.629	0.910	0.950	+0.079	+0.046	+0.026	9.65	2.96	4.12	0.0001	0.116	0.004
Trim 2.5/97.5	0.824	0.846	0.891	+0.096	+0.055	+0.014	11.16	4.05	4.82	0.067	0.188	0.000
$\bar{\sigma}^2$ lags 2–3	0.649	0.951	0.876	+0.077	+0.050	+0.020	9.85	2.87	4.44	0.000	0.080	0.004
$\bar{\sigma}^2$ lags 4–5	0.590	0.899	0.948	+0.104	+0.046	+0.021	8.94	3.01	4.38	0.000	0.063	0.040
Shock lags 0–1	0.681	0.936	0.940	+0.084	+0.016	+0.012	9.67	4.42	4.89	0.013	0.061	0.012
Shock lags 2–4	0.672	0.908	0.948	+0.098	+0.040	+0.027	8.67	3.21	4.10	0.002	0.010	0.016
Exclude 2020–21	0.642	0.992	0.886	+0.140	+0.043	+0.013	6.40	3.13	4.88	0.021	0.052	0.637

Table 7: One-at-a-time variations around the canonical specification. Bold p -values reject cross-group equality at the 5% level.

The bottom's trend. The bottom series drifts upwards (+0.83 per year, $t = 3.7$), which strains the mean stationarity behind the Blundell-Bond level moments, though the belief block does not rest on them. A linear detrend or calendar-year effects leave the bottom prior at 9.74 and 9.57 against the canonical 9.65, the bottom gain at 0.62 and 0.64 against 0.63, the gain Wald below 0.0001, and the prior Wald at most 0.008. The difference-in-Hansen statistic on the level moments does not reject ($p = 0.55$ and 0.44 at the bottom and top), and dropping them preserves the point estimates, with the bottom prior at 9.22 (1.81), at the cost of the gain's precision.

C.3 The Sign of News

Positive wage news carries roughly twice the belief weight of negative news; I treat the asymmetry as suggestive rather than established. Pooling the groups for power and splitting the shock by sign, both components load positively, with $\lambda^+ = 0.055$ versus $\lambda^- = 0.027$, significant across cluster choices ($N = 20,363$, Hansen J $p = 0.11$). The direction is the opposite of what loss aversion would predict (Kahneman and Tversky, 1979): upside news is read as informative about future earning power, while downside news is more often dismissed as transitory. Two caveats keep the reading suggestive: the extra weight may reflect the size of the surprise rather than its sign, since expected wage growth was positive over the sample, and pooling leaves open whether the asymmetry itself differs by income.

C.4 Numeracy and Inflation Uncertainty

The level gradient of Section 3 survives conditioning on numeracy and on inflation uncertainty. The SCE's numeracy module classifies 47.0 percent of bottom-group respondents as low-numeracy, against 28.4 percent in the middle and 11.1 at the top. On the full matched panel, a broader sample than the estimation panel, the bottom-to-top difference in perceived variance is 10.6 (1.0); it falls to 2.4 (0.8, $p = 0.002$) among high-numeracy respondents and rises to 17.0 (2.1) among low-numeracy ones.

Inflation uncertainty is itself steeply graded, 27.9/15.2/8.5 across the groups, and controlling for it absorbs about seventy percent of the bottom-to-top wage-risk gap, leaving a wage-specific gradient of +3.09 (0.60). On the over-control reading of Section 3, both controls remove part of the reading-noise channel itself and the conditioned gaps are lower bounds; read instead as

mechanical reporting width, they are the cleaner estimates. The gradient survives either reading, with its size bracketed by the two.

Re-estimating the belief block (16) within high-numeracy respondents reproduces the middle and the top, with priors of 2.35 (0.85) and 4.93 (1.02), at or below their realised values as in the baseline. The bottom retains only 75 estimation observations under the numeracy-module overlap, too few for the system GMM, so its prior excess cannot be checked at the estimation level.

Three checks bound the part reporting mechanics can carry. First, bottom respondents use 5.7 of the ten histogram bins, against 4.9 in the middle and 4.6 at the top, and restricting to reports that use three or more bins leaves the gradient at 23.0/12.3/12.2. Second, low numeracy widens the reported inflation density by more than the wage density in every group, with wage-against-inflation log-variance gaps of 0.92 against 1.17 at the bottom, 0.47 against 0.76 in the middle, and 0.15 against 0.62 at the top: the widening is a generic reporting trait, not a feature of wage risk.

Third, under the survey's density-implied variance, which is a parametric fit in the tradition of Engelberg et al. (2009), the group means are 28.6/16.4/11.3 against the midpoint measure's 20.8/11.2/11.3 on the same reports of the broader panel, so the gradient is steeper. The comparison qualifies Section 3: the middle-top equality there is a property of the midpoint measure, while the ordering and the bottom's excess survive under either measure.

C.5 Persistence of the Realised Volatility Process

Realised volatility is more persistent at the top than at the bottom, so the belief-persistence gradient of Table 1 runs opposite to the process it would have to be inherited from. I estimate the realised counterpart on the same estimator as the belief AR(1), a Blundell-Bond system GMM ARCH(1) in the squared wage change at lags 3–4 with two-way clustered standard errors, so the two sides differ in what they measure and not in how I measure them.

Group	Monthly		Quarterly	
	Perceived $\hat{\phi}_g$	Realised $\hat{\alpha}_{1,g}$	Perceived $\hat{\phi}_g$	Realised $\hat{\alpha}_{1,g}$
Bot	+0.343 (0.046)	+0.181 (0.014)	+0.632 (0.062)	+0.010 (0.018)
Mid	+0.142 (0.026)	+0.196 (0.011)	+0.594 (0.088)	+0.028 (0.020)
Top	+0.144 (0.025)	+0.234 (0.013)	+0.377 (0.076)	+0.058 (0.016)
Wald ($B = T$)	$p = 0.0001$	$p = 0.006$	$p = 0.009$	$p = 0.050$

Table 8: Perceived-variance AR(1) persistence (SCE) against realised wage-volatility ARCH(1) persistence (SIPP), by income group. Both come from Blundell-Bond system GMM on the same instrument set, lags 3–4 in differences plus the BB level moment, collapsed, with two-way clustered standard errors in parentheses. The realised series is stripped of the negative first-order autocorrelation that differencing mechanically induces. Respondents number 732/1,384/1,676 on the SCE side and 28,262/43,743/55,162 on the SIPP side at monthly cadence.

The monthly realised gradient rises from 0.181 at the bottom to 0.234 at the top and rejects equality of the two at $p = 0.006$, where the perceived gradient falls from 0.343 to 0.144 and rejects at $p = 0.0001$. The quarterly comparison runs the same way, though the realised coefficients are small there and individually insignificant below the top.

A cell-level check on the raw series underlying the shock regressor agrees. The within-cell monthly AR(1) is 0.29 at the bottom against 0.36 at the top, and 0.44 against 0.49 in the national series, while instrumenting the lag to remove finite-cell sampling noise drives the bottom to 0.06 against the top's 0.26. The uninstrumented numbers flatter the bottom if anything, since adjacent squared changes share a mechanical correlation through the same person-month, and that correlation is larger where cells are smaller, as they are at the bottom.

C.6 The Realised Process: Higher-Order Moments and the Stayer Conditioning

The higher-order autocovariances of the wage change are economically second order, so the permanent-transitory model behind Table 2 is an adequate description of the realised process. The model implies $\text{Cov}(\Delta e_t, \Delta e_{t-k}) = 0$ for all $k \geq 2$, and at these sample sizes the estimated moments are statistically distinguishable from zero but small, between 2 and 23 percent of the lag-1 moment that identifies $\sigma_{\theta,g}^2$. The largest deviations sit at the quarterly lags for the top, +0.99 and +0.41 at lags three and six, the signature of quarterly bonus delivery (Section 5).

The stayer conditioning does not select the sample in a way that could understate the bottom's transitory risk. Months immediately preceding an employer change carry elevated volatility at the bottom, about 1.6 times other months, but they are one percent of the sample, so excluding them moves $\sigma_{\theta,g}^2$ by less than 0.04 (%)² in every group. Closing the bottom's prior gap through selective separation would instead require the conditioning to hide a fourfold understatement of its transitory risk, far beyond the observed elevation.

The sort itself is a choice the realised process is robust to. Sorting workers by their own annual earnings rather than by household income reassigns about half of them, since spousal income moves many workers across the household cut-offs, and it leaves the pattern of Table 2 intact: the transitory variance rises more steeply with income, at 2.28, 3.49, and 8.11 (%)² across own-earnings terciles, and the permanent variance stays roughly flat, at 12.44, 11.30, and 10.86. The realised gradient therefore runs opposite to the perceived gradient under either sort. The belief side cannot be re-sorted, because the SCE elicits household income alone, so the comparison across surveys keeps the household sort.

C.7 The Same-Hours Counterpart of the Realised Process

The SCE question holds hours fixed, while the recorded SIPP wage carries them: earnings are divided by the average weekly hours reported for the month, premium pay and tips enter the numerator, and the reported schedule sets the divisor. The counterpart the conditioning calls for therefore restricts the wage changes to months in which the reported hours do not change. I recompute the moments of Table 2 keeping only consecutive within-spell changes across which the reported average weekly hours are unchanged, requiring both differences of each autocovariance pair to qualify and applying the full-sample trim bounds, so that the restriction and not the trim moves the estimates; within the restricted pairs the wage change is the change in earnings at a fixed divisor, which also frees it of the division bias the baseline carries. The restricted moments bound the object the question asks about from above rather than measure it, since reported hours are heaped, recorded changes undercount true ones, and hours worked still vary around an unchanged report.

The restriction is mild, dropping 2.3, 1.8, and 1.4 percent of observations from the bottom up, so selection on the realised outcome has little room to operate, though the dropped share itself falls with income, one more form of the exposure gradient; what it removes is concentrated all the same, thirty-one percent of the total monthly variance at the bottom against sixteen at the top. A placebo that drops the same share of changes at random leaves the permanent variance within $0.13 (\%)^2$ of its baseline in every group, so it is the months removed, not the observations lost, that carry the change.

Group	Baseline		Same reported hours		Share kept
	$\sigma_{\theta,g}^2$	$\sigma_{\psi,g}^2$	$\sigma_{\theta,g}^2$	$\sigma_{\psi,g}^2$	
Bot	2.52	11.92	2.83	6.12	0.977
Mid	4.42	11.61	4.73	6.31	0.982
Top	5.83	11.23	6.07	7.13	0.986

Table 9: SIPP permanent-transitory identification with and without the same-hours restriction, monthly cadence, in $(\%)^2$. The baseline columns re-estimate the construction of Table 2 on the same code path, matching its moments to within 0.05; the same-hours columns keep only within-spell changes across which the reported average weekly hours are unchanged, with both differences of each autocovariance pair required to qualify. Share kept is the fraction of baseline observations surviving the restriction.

Table 9 sets the restricted moments against the baseline. The transitory variance is essentially unchanged and still rises with income, so the realised gradient runs opposite to the perceived gradient under either conditioning. The netting floors of Appendix D.3 move to 2.49, 3.89, and 3.90, the middle and top priors remain within one standard error of them, and the bottom's excess is 7.16 ($t = 4.1$) against 7.43 at the baseline; rescaling the labelled share to the restricted movement, which removes hours-event variance from its denominator, raises the shares to about 0.17, 0.23, and 0.43 and moves none of the three comparisons. The full-information attribution ratio $\lambda_{\text{Bot}}^* / \lambda_{\text{Top}}^*$ moves from 1.43 to 1.40, and the closed form (10) at the restricted moments predicts attribution ratios of 1.71 and 1.25 in place of 1.69 and 1.31, still within one standard error of the estimates; the persistence comparison of C.5 is equally insensitive, showing no bottom-top gap whether the wage is measured per reported hour or raw.

The permanent variance falls by roughly half at the bottom and the middle and by over a third at the top. A change in the reported schedule is rare but persistent. It is a move between full- and part-time work or a renegotiated contract rather than a noisy month, so the moments of Section 3 assign what it carries to the permanent component; the hours-linked permanent share, 5.8, 5.3, and $4.1 (\%)^2$ from the bottom up, itself falls with income, showing the delivery gradient of Section 5 on one more margin. The twelve-month benchmark (5) falls to 79, 85, and $98 (\%)^2$ and now rises with income where the reported variance falls, so the correction does not cancel from the cross-group comparison; it strengthens the contrast, and the reported variance sits about four to nine times below the tighter benchmark in every group, so the shortfall on which the constant-attribution evidence of Section 4 rests is preserved.

The removed variance also gives the bottom's prior excess a second possible source. A bottom worker unable to comply with the instruction, who folded part of the hours-event risk into her answer, would be reporting truthfully and would still sit above her netting floor, and folding in

about a fifth of the removed variance would account for the excess in full. The moments cannot separate this partial inclusion from unresolved reading of the pay she does condition on, while the inflation-density widening of Appendix C.4, which no hours channel reaches, indicates that part of the excess is generic reporting rather than either. Little turns on the allocation for behaviour, since each reading enters the transitory prior and not the permanent belief, but the excess is best described as reported risk beyond the object the question asks about, with reading noise the component the mechanism predicts rather than the whole the data establish.

The baseline moments remain those of Table 2, because the budget of Section 6 must carry the earning-power risk the household actually absorbs, and a persistent cut in scheduled hours is exactly such risk; the question conditions it out of the elicited object, not out of her income. The cost of the choice is bounded in closed form: at the restricted moments and the measured shares the resting discounts $V_u/\bar{S}_m$ deepen to 0.94, 0.89, and 0.77 while the twelve-month permanent gaps shrink from -5.0 , -11.9 , and -25.3 to -4.3 , -8.1 , and -19.3 (%)², so the wedge of Section 6 keeps its shape and its concentration at the top at magnitudes smaller by between a seventh and a third.

C.8 The Benchmark at Long Horizons

The twelve-month benchmark of Section 3 extrapolates monthly moments under the random walk rather than measuring the twelve-month change, so this subsection measures it directly on the same panel. For each horizon h I take within-spell pairs h months apart, drop the pair whose later endpoint is January as the monthly construction does, and trim the squared change at the same one and ninety-nine percent. At $h = 1$ the exercise returns the moments of Table 2 by construction, 16.9, 20.4, and 22.9 (%)². At $h = 12$ it returns 2724, 2610, and 2774, about eighteen times the extrapolated benchmark.

The gap is a statement about outliers rather than about the wage process. A job stayer's monthly change is a spike at zero with heavy tails, since most months carry no change in the reported rate, so the mean-based moment is governed by the tail and the common trim removes about eighty-five percent of it at one month against under half at twelve. Measuring the same profile with a scale that ignores the tails, the interquartile range of the change in Gaussian variance units, moves the comparison the other way: it puts the monthly variance near 0.6, far below the trimmed moment, and the twelve-month variance near 340, twice the benchmark rather than eighteen times it. One measure is governed by the tail and the other by the spike, and the data do not discipline the choice between them.

The level of the benchmark is therefore not identified without a model of the reporting outliers, which is why Section 3 reads it as an upper bound and rests the argument on the contrast across groups rather than on the level. What the profile does settle is the direction: the directly measured twelve-month variance lies above the extrapolation under either scale, so the shortfall of reported risk against realised risk is not an artefact of the random-walk extrapolation.

D Identification: the Filter, the Labelled Share, and the Scorecard

This appendix gives the filter behind the gain (D.1), the measurement of the labelled share and its robustness (D.2), the scorecard setting the measured mechanism against the estimated belief moments (D.3), the filter's diagnostic on the estimated gains (D.4), and the hours evidence behind the reading-noise gradient (D.5).

D.1 The Filter

Once the attribution is rescaled to the agent's own signal, the perceived transitory prior drops out of the filter's signal-to-noise ratio, so the gain depends on the labelled share and not on reading noise.

The agent's monthly signal is her own squared netted reading, $x_t = (\Delta \tilde{y}_t)^2$ with $\tilde{y}$ from (7). It has mean $\sigma_{\psi,g}^2 + V_{x,g}$ and, under Gaussianity, variance $2(\sigma_{\psi,g}^2 + V_{x,g})^2$, where $V_{x,g} = 2\sigma_{\theta,\text{perc},g}^2$ is twice the perceived transitory prior. The perceived permanent variance follows a random walk with innovation variance q (the drift), the filter's one free parameter, fitted to the estimated gains in D.4. The observation is the attributed signal

$$z_t = \tilde{\lambda}_g x_t = \hat{\sigma}_{\psi,t}^2 + \varepsilon_t,$$

a unit-loading reading of the state, where $\tilde{\lambda}_g = \sigma_{\psi,g}^2 V_{u,g} / [\bar{S}_{m,g}(\sigma_{\psi,g}^2 + V_{x,g})]$ is the full-change attribution (10) rescaled to the signal's mean and $V_{u,g} = \sigma_{\psi,g}^2 + 2(1 - \rho_g)\sigma_{\theta,g}^2$ is the unlabelled residual variance of the main text. The rescaling makes the observation unbiased at rest, since $\tilde{\lambda}_g \mathbb{E}[x_t] = \sigma_{\psi,g}^2 V_{u,g} / \bar{S}_{m,g}$ is the permanent belief (11). The variance of a squared signal moves with the state, so I evaluate it at the resting point and read the filter as moment-matched rather than exact. Its steady-state gain is

$$\mu_g = \frac{\bar{P}_g}{\bar{P}_g + R_g}, \quad \bar{P}_g = \frac{1}{2}(q + \sqrt{q^2 + 4qR_g}),$$

with $\bar{P}_g$ the positive root of the Riccati equation, and the observation noise is

$$R_g = \text{Var}(z_t) = (\tilde{\lambda}_g)^2 2(\sigma_{\psi,g}^2 + V_{x,g})^2 = 2(\sigma_{\psi,g}^2)^2 (V_{u,g} / \bar{S}_{m,g})^2.$$

The prior enters the attribution and the signal variance by offsetting factors and cancels from R_g , so the gain depends on the labelled share, through $V_{u,g}$, and not on reading noise. The model carries the gain as the estimated moment, and the filter serves only as the D.4 diagnostic on the gains' ordering.

D.2 Measuring the Labelled Share

The labelled share of within-spell pay movement rises with income, from 0.121 at the bottom through 0.178 in the middle to 0.358 at the top, and the gradient survives every specification in Table 10. I measure it on the stayer panel of Section 2: 2,250,815 person-months across 103,134 persons and 127,184 spells.

For a worker i in job spell s , let y_{it} be total earnings in month t and ℓ_{it} labelled pay, bonuses and commissions in the main specification, with spell means $\bar{y}_s$ and $\bar{\ell}_s$. I demean each within the spell and scale by spell-mean earnings,

$$d_{it}^{\text{tot}} = \frac{y_{it} - \bar{y}_s}{\bar{y}_s}, \quad d_{it}^{\text{lab}} = \frac{\ell_{it} - \bar{\ell}_s}{\bar{y}_s},$$

and take the person-weighted coefficient from regressing the labelled deviation on the total through the origin,

$$\rho_g = \frac{\sum_{(i,t) \in g} w_{it} d_{it}^{\text{lab}} d_{it}^{\text{tot}}}{\sum_{(i,t) \in g} w_{it} (d_{it}^{\text{tot}})^2} = \frac{\text{Cov}_w(d^{\text{lab}}, d^{\text{tot}})}{\text{Var}_w(d^{\text{tot}})},$$

where w_{it} are the SIPP person weights and the sum runs over stayer person-months in group g . Seam months are dropped as in Section 2.

Writing $d^{\text{tot}} = d^{\text{lab}} + d^{\text{unlab}}$ gives $\rho_g = [\text{Var}_w(d^{\text{lab}}) + \text{Cov}_w(d^{\text{lab}}, d^{\text{unlab}})] / \text{Var}_w(d^{\text{tot}})$, which reduces to the pure variance share when the two parts are uncorrelated; it appears as the alternative row of Table 10 and stays close to the main row throughout. The denominator carries permanent drift and any tenure trend, so ρ_g is a conservative proxy for the labelled share of $\sigma_{\theta,g}^2$ alone.

Specification	Bot	Mid	Top
Main: within-spell demeaned, labelled = bonus + commission	0.121	0.178	0.358
Variance share instead of regression share	0.146	0.203	0.373
Within-spell detrended	0.176	0.256	0.488
Imputed component months dropped	0.042	0.085	0.314
Overtime counted as labelled	0.134	0.197	0.370
Annual spell-year cells (different frequency)	0.050	0.060	0.075

Table 10: The measured labelled share ρ_g across specifications, by income group, SIPP stayer panel 2013–2023, person-weighted. The main row is the central specification, and the model pins its share within the band these rows span (Section 6).

The remaining rows set the measurement band. The detrended row removes a within-spell linear trend that demeaning leaves in the unlabelled residual and bounds the share from above. The imputation row bounds it from below: imputation that spreads labelled dollars across months can manufacture labelled variance, so the row drops every imputed person-month, lowering the shares but leaving the gradient intact; it is the band’s lower edge, since it also discards genuine labelled months. The annual row does not enter the band: at spell-year cells the monthly variation the filter reads has averaged out, so it measures the labelled fraction of year-to-year risk.

Receipt shares in the same records confirm the direction of the delivery gradient. Bonus receipt rises from 2.8 to 4.8 to 9.7 percent across the groups in 2018 (2.8/7.1/12.1 in 2021), receipt of any labelled component from 7.9 to 16.5 percent between bottom and top, and receipt of tips falls from 2.4 to 0.9. Receipt speaks to direction, not the level of ρ , and places the middle below the top on every labelled component.

A final check: the netting floor $(1 - \rho_g)\sigma_{\theta,g}^2$ that D.3 scores the priors against multiplies a level-based share by the trimmed difference moment of Section 3. Measured directly as the lag-1

autocovariance of the unlabelled pay change, the floor moves by about a tenth or less under the trim in every group, while the total difference moment moves by factors of seven to fifty, because the trimmed tail is almost entirely labelled bonus and commission months. The labelled share implied by the difference moments, $0.08/0.10/0.48$, lies inside the band at the bottom and the middle and at its detrended upper edge at the top, so the top's floor comparison supports the band rather than the point.

D.3 The Scorecard

With the labelled share measured, the estimated belief moments of Section 5 become out-of-sample predictions, scored by three comparisons. The floors isolate rather than predict a fourth number: the bottom's residual.

The first comparison sets the estimated priors against the netting floors $(1 - \rho_g)\sigma_{\theta,g}^2 = 2.21/3.63/3.74$. The middle and top priors sit on them: the residuals $\sigma_{\theta,\text{perc},g}^2 - (1 - \rho_g)\sigma_{\theta,g}^2 = \hat{\omega}_g^2$ of (13) are -0.68 and $+0.38$, with standard errors 0.94 and 0.82 , both within one standard error of zero.

The middle's estimate sits 0.7 standard errors below its floor, an insignificant violation of the inequality $\omega^2 \geq 0$; the model carries the censored value. The bottom's residual, $+7.43$ (standard error 1.76 , $t = 4.2$), is the identification of the reading noise, concentrated where hours volatility is highest (D.5). The comparisons repeat under the same-hours moments of Appendix C.7, with residuals of -0.93 (0.94) at the middle, $+0.22$ (0.82) at the top, and $+7.16$ (1.76) at the bottom.

Reporting error in either survey runs against the bottom's residual rather than for it. Classical error in SIPP earnings loads one-for-one into measured $\sigma_{\theta,g}^2$ and hence into the floors, so at an error share s of transitory variance each residual rises by $s\sigma_{\theta,g}^2$: the middle's stays within one standard error of zero for s up to 0.37 and insignificant up to 0.57 , the top's crosses one standard error at $s = 0.08$ and significance at 0.21 , and the bottom's excess only grows.

On the SCE side, a common understatement of uncertainty would deflate every prior and understate the excess further, while an overstatement large enough to manufacture it, a factor near four, would push the middle and top priors far below floors they instead sit on. The group-specific overstatement that remains is the numeracy channel, conditioned on in Appendix C.4.

The second comparison runs on cross-group attribution ratios rather than the level, because the cell-mean shock regressor carries a worker's own shock only in part, so the slope recovers her response up to a loading the merge itself does not pin down. The loading can be measured in the SIPP: regressing each worker's own trimmed $(\Delta e_{i,t})^2$ on the cell mean of the other workers in her cell, constructed exactly as the regressor is, gives 0.43 , 0.56 , and 0.58 from the bottom up.⁹ The loadings are similar though not identical, and the difference runs in the conservative direction: correcting the ratios by them raises bottom-to-top from 3.04 to about 4.1 and moves middle-to-top

⁹The projection is the least-squares loading, which approximates the factor the system estimator recovers through lagged instruments. Its common variation is in large part the seasonal raise cycle, which under the rule is genuine updating variation, the realised squared changes the agent responds to whether or not their season is predictable; stripping it, by deseasonalising the cell series by month-of-year or by pushing the instruments beyond the smoothing window, weakens all three estimated responses and degrades the estimator's moment conditions, so the news response is identified largely off the seasonal arrival of wage news.

from 1.76 to 1.83, so treating the loading as common understates the gradient the mechanism is asked to explain rather than manufacturing it.

At the measured shares the closed form (10) predicts bottom-to-top and middle-to-top attribution ratios of 1.69 and 1.31 (in logs 0.52 and 0.27), against estimated ratios of 3.04 and 1.76 (in logs 1.11 and 0.57, with log standard errors 0.63 and 0.59). The gaps are $z = +0.9$ and $+0.5$, within one standard error, and the predictions are about half of the estimated point gradient. Under the measured-loading correction the estimated gradient widens and the mechanism's share falls to about a third at the bottom; I keep the uncorrected comparison as the baseline, since the loading is itself an approximation, and read the corrected comparison as the outer edge. The moment has little power against full information: the FIRE gradient $\lambda_{\text{Bot}}^*/\lambda_{\text{Top}}^* = 1.43$ (log 0.36) sits $z = 1.2$ from the estimate, so the ratios discipline the mechanism's direction and rough size, not its existence.

The third comparison is the gain diagnostic of D.4, which predicts the gains' pinned contrast, with the bottom slowest, and underpredicts their spread, on the moments the cadence comparison of Section 3 marks as the least reliable. The tally is two moment sets consistent within one standard error, one ordering the filter gets right while underpredicting its spread, and one significant residual the mechanism names rather than predicts.

D.4 The Gain Diagnostic

The filter of D.1, run at the measured shares, predicts the ordering of the estimated gains but underpredicts their spread. The observation noise is $R_g = 2(\sigma_{\psi,g}^2)^2(V_{u,g}/\bar{S}_{m,g})^2 = 262.7/229.3/168.3$, which falls with the labelled share, so any common drift q implies a gain that rises with income. At the common drift that best matches the three estimated gains, weighted by their precisions, $q^* = 646.5$, the predicted gains are 0.763/0.783/0.823 against the estimated 0.629/0.910/0.950. The Gaussian factor of two in R_g is not load-bearing: the kurtosis of the trimmed residualised wage change is heavy but nearly common across groups (56.5/54.1/52.3), so the empirical observation noise rescales the three groups together, and the refitted drift absorbs the common factor, leaving the predicted ordering and the spread shortfall unchanged.

The diagnostic also separates the constant attribution from the Bayesian rescaling of Section 4. The rescaler's attributed signal has variance $2(\sigma_{\psi,g}^2)^2 = 282.7/269.1/251.8$, so any common drift implies a nearly flat gain profile, at 0.79/0.79/0.80 under the same fit, against the estimated rise from 0.63 to 0.95. The noise of the constant-attribution filter falls with the labelled share instead, and it predicts the rising order. The attribution itself cannot settle the matter, because the two rules imply coefficients on the observed squared change that differ only by the factor $V_{u,g}/\bar{S}_{m,g}$, well within the standard errors (Appendix B). The evidence is therefore consistent with that reading rather than decisive for it, since a Bayesian free to set her drift group by group can match any profile of gains.

Experience-based learning grades updating by age (Malmendier and Nagel, 2011), so an income gradient in the gain could be an age gradient in disguise. Within the 40-plus age band, however, the estimated gain rises from 0.57 to 0.84 to 0.98 across the income groups, and the bottom-to-top gap is over four standard errors. The under-40 band retains too few bottom-group respondents (117) to estimate the gain, so the check rests on the older band alone.

D.5 Hours by Income Group

Within-person hours volatility $\mathbb{E}[(\Delta \log h)^2]$ falls monotonically with income, 4.10/3.06/2.36 in $(\%)^2$ (Figure 10), which is the direction the reading-noise residual of D.3 requires. The reported hours variable is heaped, so nearly all of this volatility comes from the rare schedule changes that Appendix C.7 nets out of the realised process; the gradient therefore measures a worker's exposure to schedule events, and it bears on the reading problem as a proxy, since the within-schedule movement she must read through is exactly what a heaped report cannot record. Transitory shocks delivered through hours are also of the unlabelled kind, so variable hours lower ρ_g as well. The earnings stream itself is lumpiest at the bottom: the kurtosis of the trimmed raw log earnings change falls from 46.8 through 24.1 to 15.5 across the groups, where the residualised wage change of D.4 is nearly common, so the flow she must read is more leptokurtic as well as more volatile. The check is directional only: a proportional scaling of the residual in hours volatility, calibrated at the bottom (7.43/4.10), would predict a middle residual near 5.5 against an estimated -0.68 (standard error 0.94), so no intensity mapping from observables to ω^2 is estimable.

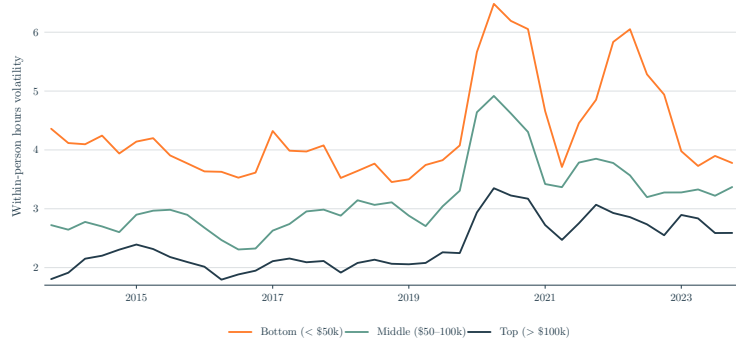


Figure 10: Within-person hours volatility $\mathbb{E}[(\Delta \log h)^2 | g, t]$ in $(\%)^2$ by income group. SIPP matched-stayer panel, quarterly with 4-quarter trailing MA.

E Calibrating the Belief Profile

This appendix builds the continuous belief profile the model consumes and defines the upper-bound variant that Section 6 reports.

E.1 From Three Groups to a Continuum

The terciles are the finest income split the SCE powers, so the profiles of Figure 5 interpolate the three groups rather than estimate finer bins. Re-estimating on the five native SCE brackets breaks the belief block where it is split: the lowest bracket (under \$20k, 102 respondents) has no estimable cell, the attribution turns negative in the lower-middle, the gain exceeds one in the top two, and only the middle, the unchanged tercile, survives.

Interpolation. The share ρ and the gain μ interpolate in levels and the prior $\sigma_{\theta, \text{perc}}^2$ in logs, monotone cubic through the three anchors and flat outside them. End tangents are zero, so each profile joins its flat extension without a kink; interior tangents are bounded so every segment stays monotone. The prior's floor censoring at $(1 - \rho(y)) \sigma_{\theta}^2(y)$ binds only at the middle anchor itself.

Table 11 evaluates the mapping at representative incomes. The attribution, the discount, and the permanent belief follow from the closed forms: the λ column is the estimating form (10), and $\tilde{\lambda}$ is the implemented form on the agent's own netted reading.

Monthly income	Carried ρ	Carried		Attribution		Discount $V_u/\bar{S}_m$	Perceived perm. var
		μ	$\sigma_{\theta, \text{perc}}^2$	λ	$\tilde{\lambda}$		
\$1,500	0.119	0.629	9.65	0.677	0.368	0.965	11.47
\$2,415 (Bot)	0.119	0.629	9.65	0.677	0.368	0.965	11.47
\$3,500	0.134	0.718	6.72	0.610	0.445	0.952	11.21
\$6,014 (Mid)	0.197	0.910	3.55	0.519	0.568	0.915	10.61
\$9,000	0.297	0.945	3.82	0.453	0.515	0.860	9.81
\$13,604 (Top)	0.369	0.950	4.12	0.398	0.468	0.812	9.11
\$20,000	0.369	0.950	4.12	0.398	0.468	0.812	9.11

Table 11: The income-to-belief mapping of Figure 5 evaluated at representative incomes. Rows away from the three group anchors (Bot, Mid, Top) are interpolated, not separately measured. In $(\%)^2$ except the dimensionless shares.

E.2 The Upper-Bound Variant

The belief-anchored variant takes the estimated attribution responses at face value rather than reading them through the measured delivery of pay. It anchors the bottom's attribution at its full-information share $\lambda^* = \sigma_{\psi}^2/\bar{S}_m = 0.702$ and lets the estimated attribution ratios supply the middle and the top, giving $\lambda = 0.702/0.407/0.231$; the resting permanent belief $\lambda \bar{S}_m$ is then 11.89/8.32/5.29, discounts of 1.00/0.72/0.47 on the truth. The prior is carried as in the baseline. The variant's wedge is 8.5, 19.5, and 32.4 percent of wealth and +1.2, +3.1, and +6.7 points on the marginal propensity: the shape matches the measured calibration at roughly twice the size. The gap between the two calibrations is the part of the wedge that rides on netting beyond the measured components, or on sampling noise in ratios this wide.

F Steady-State Model Output

Table 12 gives the perceived variance under the measured beliefs against FIRE, by income group.

Group	$\hat{\sigma}_{\psi}^2$	$\sigma_{\theta, \text{perc}}^2$	Perceived $\bar{\sigma}_{\text{perc}}^2$	FIRE	Ratio
Bot	11.3	8.3	153	148	1.03
Mid	10.8	5.1	139	148	0.94
Top	9.7	3.9	124	147	0.84

Table 12: Stationary perceived variance against FIRE at the 12-month horizon, in $(\%)^2$: simulated cross-sectional means by tercile under mobility. $\hat{\sigma}_{\psi}^2$ is the mean permanent belief and $\sigma_{\theta, \text{perc}}^2$ the mean transitory prior, with realised counterparts 11.8/11.6/11.4 and 2.9/4.0/5.3; $\bar{\sigma}_{\text{perc}}^2 = 12\hat{\sigma}_{\psi}^2 + 2\sigma_{\theta, \text{perc}}^2$ and $\text{FIRE} = 12\sigma_{\psi}^2 + 2\sigma_{\theta}^2$. Entries are computed from unrounded inputs, so displayed sums and ratios can differ in the last digit.

Table 13 gives the numbers behind Figure 7.

Group	Mean wealth (months)			12-month MPC		
	FIRE	Belief	Δ	FIRE	Belief	Δ
Bot	2.28	2.17	-5.1%	0.470	0.477	+0.7pp
Mid	1.71	1.57	-8.3%	0.411	0.423	+1.3pp
Top	1.50	1.29	-13.8%	0.377	0.400	+2.4pp
<i>Workers only, re-terciled among workers</i>						
Bot	2.20	2.07	-5.8%	0.334	0.343	+0.9pp
Mid	1.54	1.39	-9.7%	0.359	0.374	+1.5pp
Top	1.47	1.25	-15.1%	0.348	0.375	+2.7pp

Table 13: Belief regime versus FIRE by income tercile. Wealth is in months of permanent income, the 12-month MPC is the annualisation $1 - (1 - m)^{12}$ of the one-month response m to a transitory windfall, and the regimes are paired by common random numbers. Partial equilibrium at $r = 2$ percent with common $\beta = 0.917$, which targets mean liquid wealth of about 1.8 months. The bottom tercile is 54 percent retirees, against 22 and 12 percent above; the workers-only panel re-terciles among workers. Its MPC levels sit below the full-population levels and are not monotone, because removing retirees removes short horizons and high propensities, while the belief wedge itself sharpens.

Table 14 gives the general-equilibrium counterpart of Section 6.

Regime	r (% ann.)	K	H2M share
FIRE	3.00	3.65	0.338
Belief	3.21	3.45	0.342

Table 14: General-equilibrium FIRE versus belief, common $\beta = 0.917$, one-asset closure, with Z pinned from FIRE at $r^* = 3\%$ and held fixed across regimes. H2M is the hand-to-mouth share.